

GeoLabeller

How-To

Software version v2.9.3

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A guide to labelling imagery with GeoLabeller, in two parts: placing and managing labels on the map, and recording what each label shows in the Snippet Editor. Every picture in it was taken from the running software by the script that builds this document, so the pictures show the version named above.

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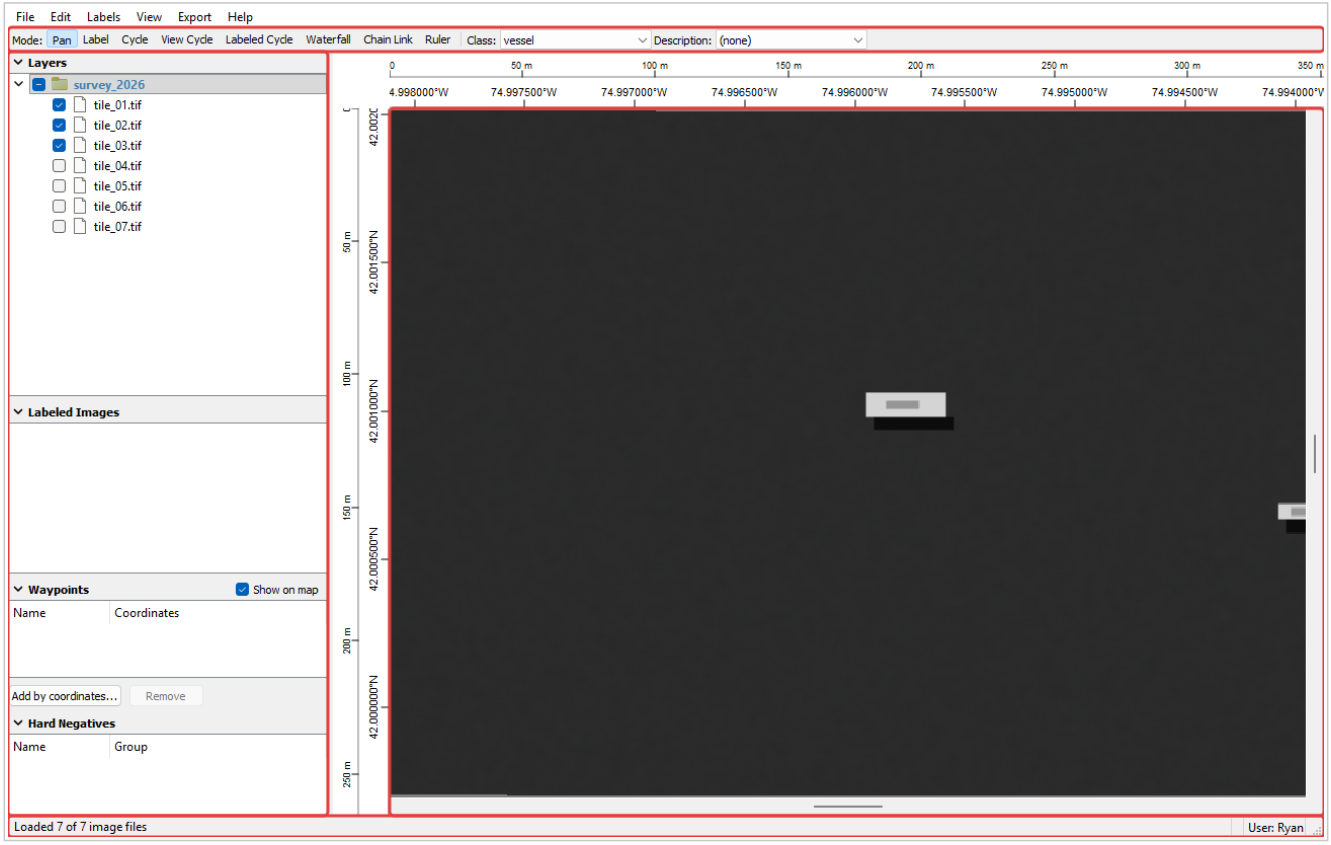
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Part 1

Labelling imagery on the map

1 The window: what each part is for

GeoLabeller opens with an empty project. Everything you do is recorded in the project, which you save to a file (File > Save Project); the imagery itself is never copied, only referred to.

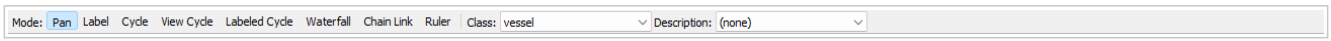


The main window with a folder of imagery loaded. Left: the panels. Top: the mode toolbar. Middle: the map. Bottom: the status bar.

The left column: Layers, Labeled Images, Waypoints, Hard Negatives

Four panels, each folding under its header. LAYERS is the imagery, by folder, with a checkbox to show or hide each image. LABELED IMAGES lists every image with a label on it, by object. WAYPOINTS are named places to return to. HARD NEGATIVES lists the images flagged as holding nothing of interest (chapter 7).

The toolbar: mode, class and description preset



The toolbar.

The MODE decides what a click on the map does - pan, place a label, step through images - and has a key each (chapter 12). CLASS is the class the next label gets; DESCRIPTION, a preset description for it (chapter 4).

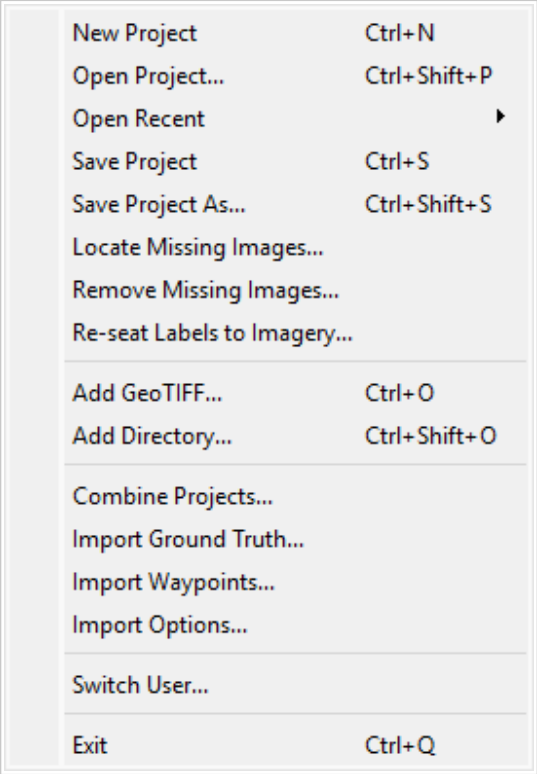
The status bar: what a click will do right now

The status bar answers 'what happens if I click now?' and reports what the last action did. Glance at it when unsure. At its right is the user name, recorded on every label, mask, rating and review you make - File > Switch User... changes it. Over an image the readout at the right also shows the pixel under the cursor: every band's value, and through a colour table the value's colour. View > Pixel Values in Status Bar turns that off.

2 Loading imagery

A folder of GeoTIFFs as a group; single files

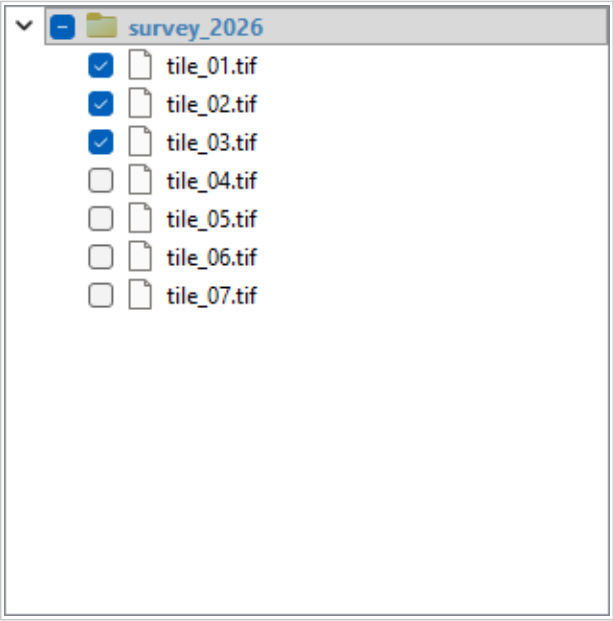
1. File > Add Directory... and choose the folder. Every GeoTIFF in it (and in its sub-folders) is loaded, as a group named after the folder.
2. File > Add GeoTIFF... loads single files instead.



The File menu.

Showing and hiding images; Zoom to layer

Nothing is shown until it is ticked, so a large survey does not fill the screen at once. Tick an image in the Layers panel to show it, or the group to show them all; right-click an image > Zoom to layer puts it in front of you.



The Layers panel: a folder loaded, three images ticked.

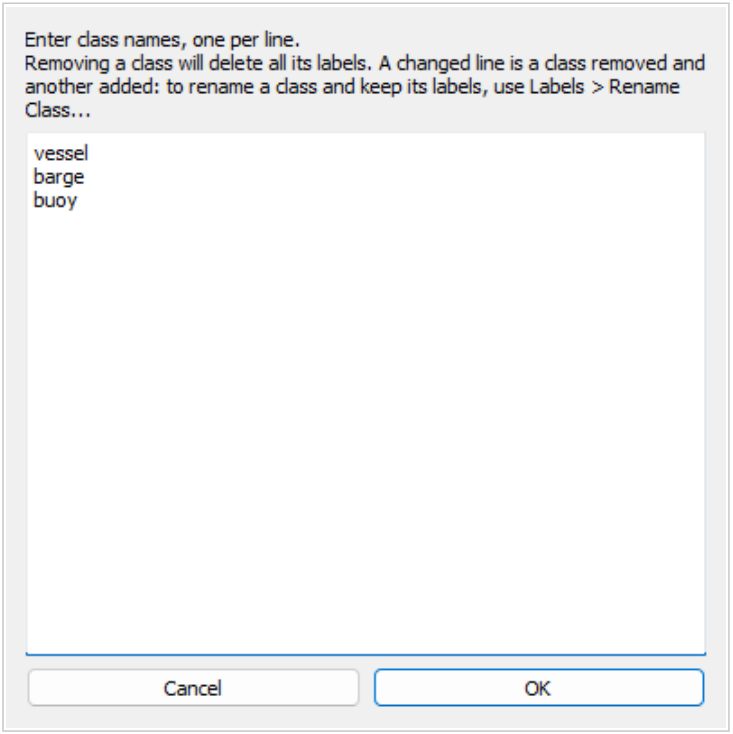
Imagery without georeferencing

Imagery that carries no georeferencing is loaded too, under a 'Non-Georeferenced' root, and laid out side by side rather than on the ground. Labels on it have pixel positions but no latitude and longitude.

> How a group is drawn - which bands, brightness, contrast and gamma - is set per group under View > Display Settings... (Ctrl+D). It changes the display only (chapter 9).

3 Setting up the classes your project records

A label has a class: what the thing is. Set the classes up once under Labels > Edit Classes..., then pick the class for the next labels from the Class box on the toolbar - or press its number: 1 for the first class in the list, 2 for the second, and so on.



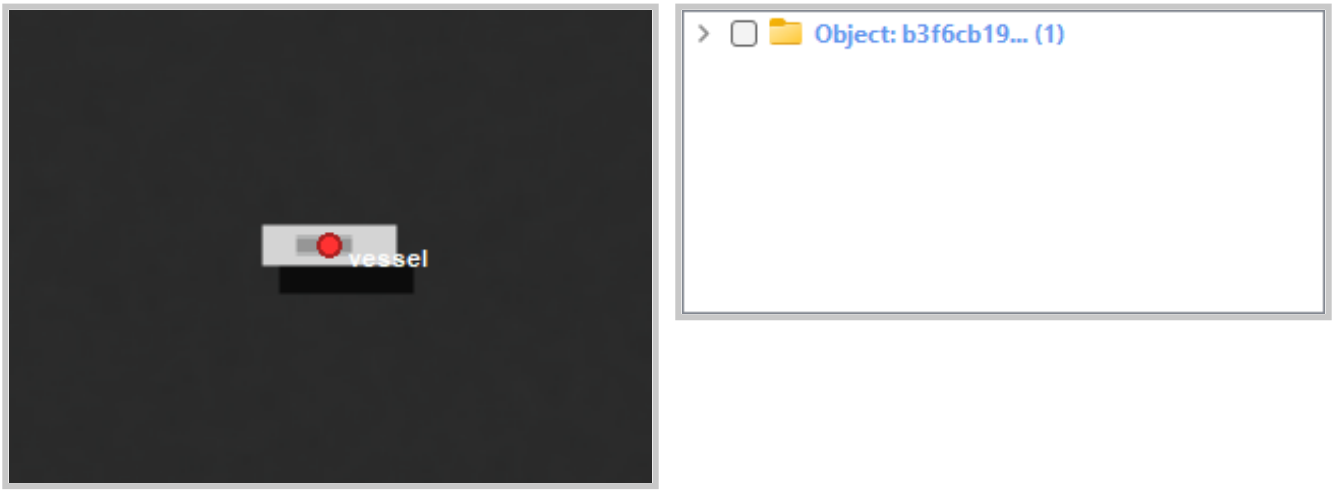
Labels > Edit Classes...

> Removing a class removes its labels; the dialog says how many before it does. The change is one undo step.

4 Placing a label on an object

Label mode and the marker

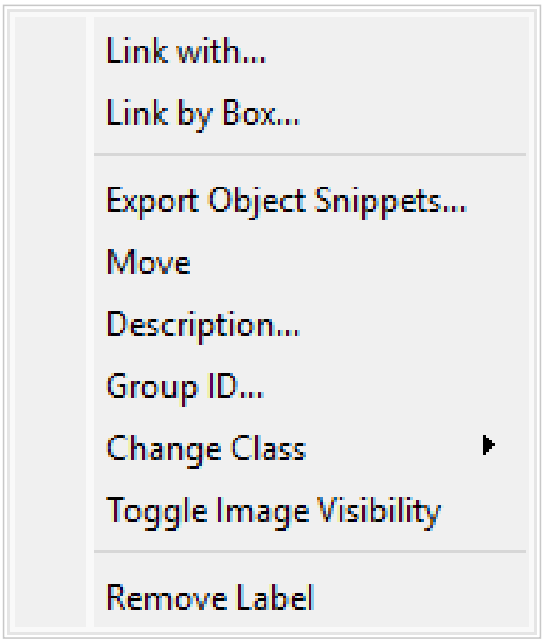
1. Press L (or click Label on the toolbar). The status bar now says a click places a label.
2. Click the object. A marker appears in the class's colour, and the image joins the Labeled Images panel.
3. Keep clicking objects. In a cycle mode (chapter 6) Space moves to the next image, so a pass over a survey is click, Space, click, Space.



A label just placed, and the Labeled Images panel listing it. The panel's checkboxes show and hide images too.

Everything a label's right-click menu offers

Right-click a label for everything that can be done to it: link it to the same object in another image (chapter 5), export its snippets, move it, describe it, name its group, change its class, hide its image, or remove it.



A label's right-click menu.

Description presets for the labels you place

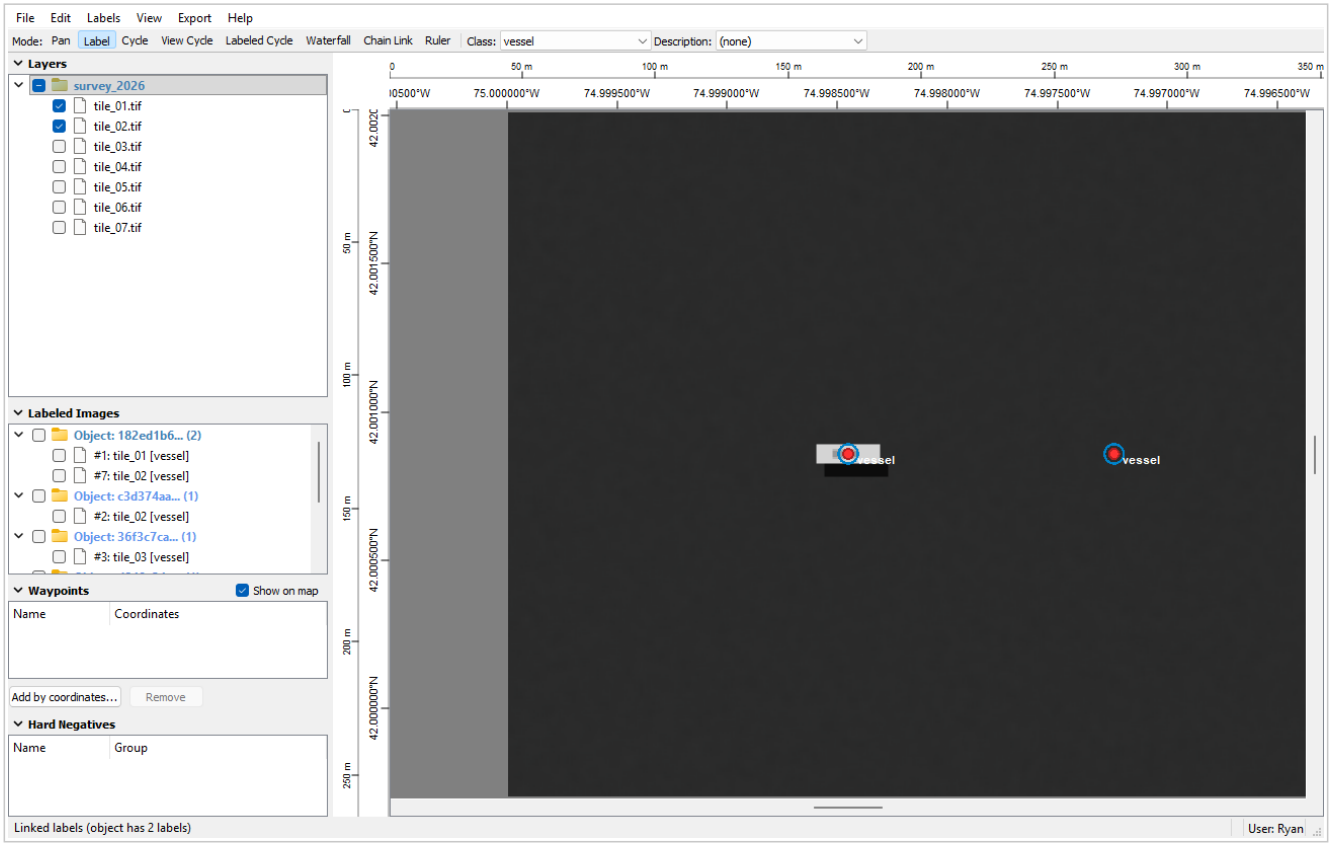
The Description box on the toolbar holds preset descriptions (Labels > Edit Descriptions...). Whichever is chosen is given to every label you place - Shift+1 to Shift+9 switch presets, Shift+0 chooses none. A description can also be typed per label, from its menu or in the Snippet Editor (chapter 21).

5 Linking the same object across overlapping images

Where surveys overlap, the same vessel appears in more than one image. Link those labels and GeoLabeller knows they are one object: it is exported once per view but counted once, a heading or size measured on one view can be given to the others, and the Snippet Editor shows every view of the object together.

Link with..., Link by Box, Chain Link

1. Right-click a label > Link with..., then click the other label.
2. Or right-click > Link by Box... and drag a box round all the labels of one object.
3. For many objects in a row, press K (Chain Link): click the labels of one object, press N to start the next, Escape to finish.
4. Linked labels wear a halo in a shared colour. Right-click > Unlink takes one out of its object.



Two overlapping images with a linked pair: the same vessel labelled in each, haloed in one colour. Labels on the image in front are bracketed.

Group IDs: one name for the whole object

A Group ID (right-click > Group ID...) is a name for the whole linked object - a hull number, say - and is given to every label linked to it. The Snippet Editor lists objects by it.

6 Stepping through a survey: the cycle modes

The cycle modes step through a group's images one at a time, showing only the current one, so a survey of hundreds of tiles can be worked through in order without hunting in the Layers panel. Select the group in the Layers panel first. In every cycle mode a left click still places a label, right-drag pans, the wheel zooms, and the status bar says where you are. Each mode suits a different job.

Cycle (C): every image in the group, in order

Space shows the next image, Ctrl+Space the previous, from the first tile to the last. Use it for the first pass over a survey, when every tile has to be looked at and nothing may be skipped: the order is the folder's, so the count on the status bar is also how far through the survey you are.



Cycle mode's status line.

View Cycle (V): only the images in the current view

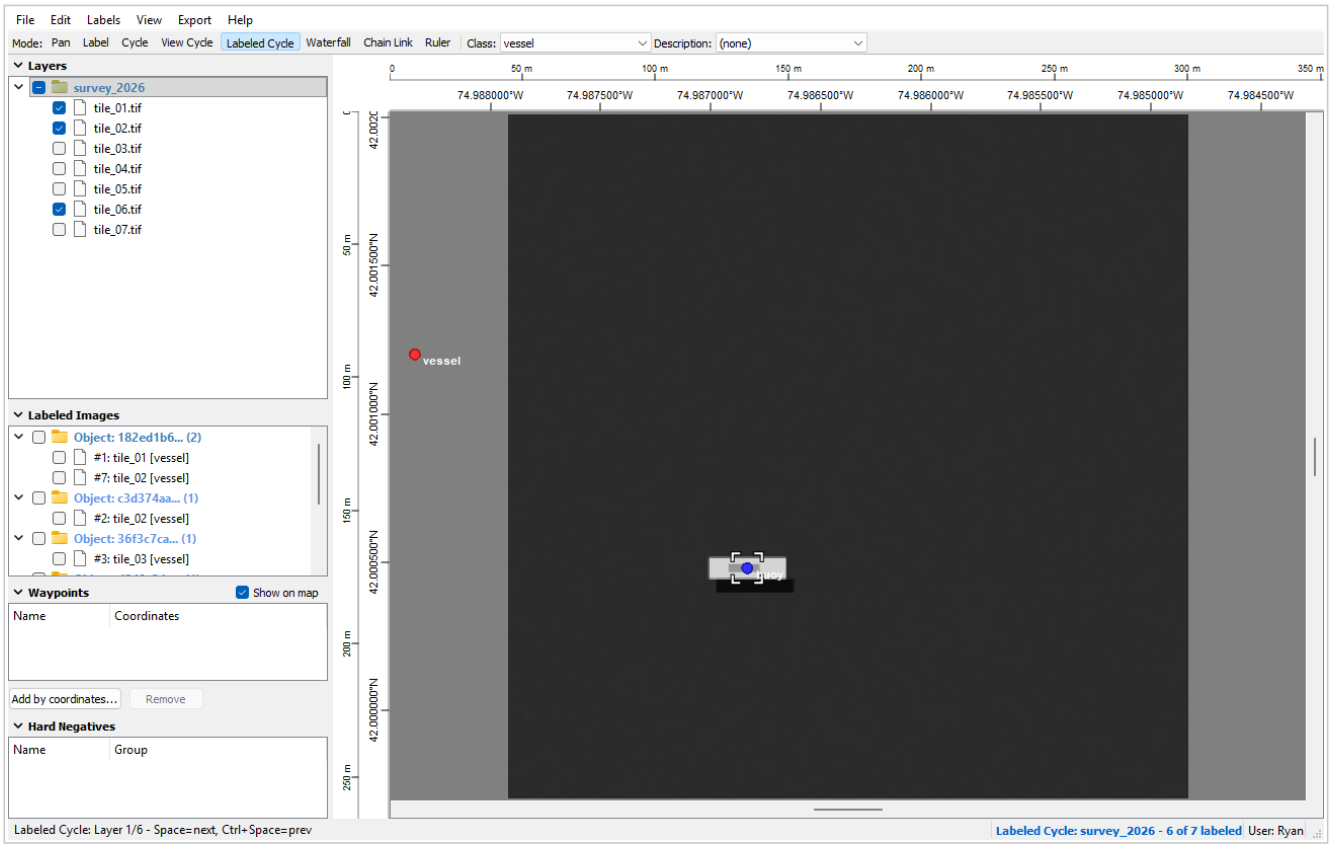
The same stepping, over just the tiles that fall within what the map is showing. Zoom the map to one area of interest - a harbour, a stretch of coast - and work it tile by tile without leaving it. Zoom out again and the cycle takes in more. The cycle remembers its place while you look elsewhere.



View Cycle's status line.

Labeled Cycle (Shift+C): only the images with labels

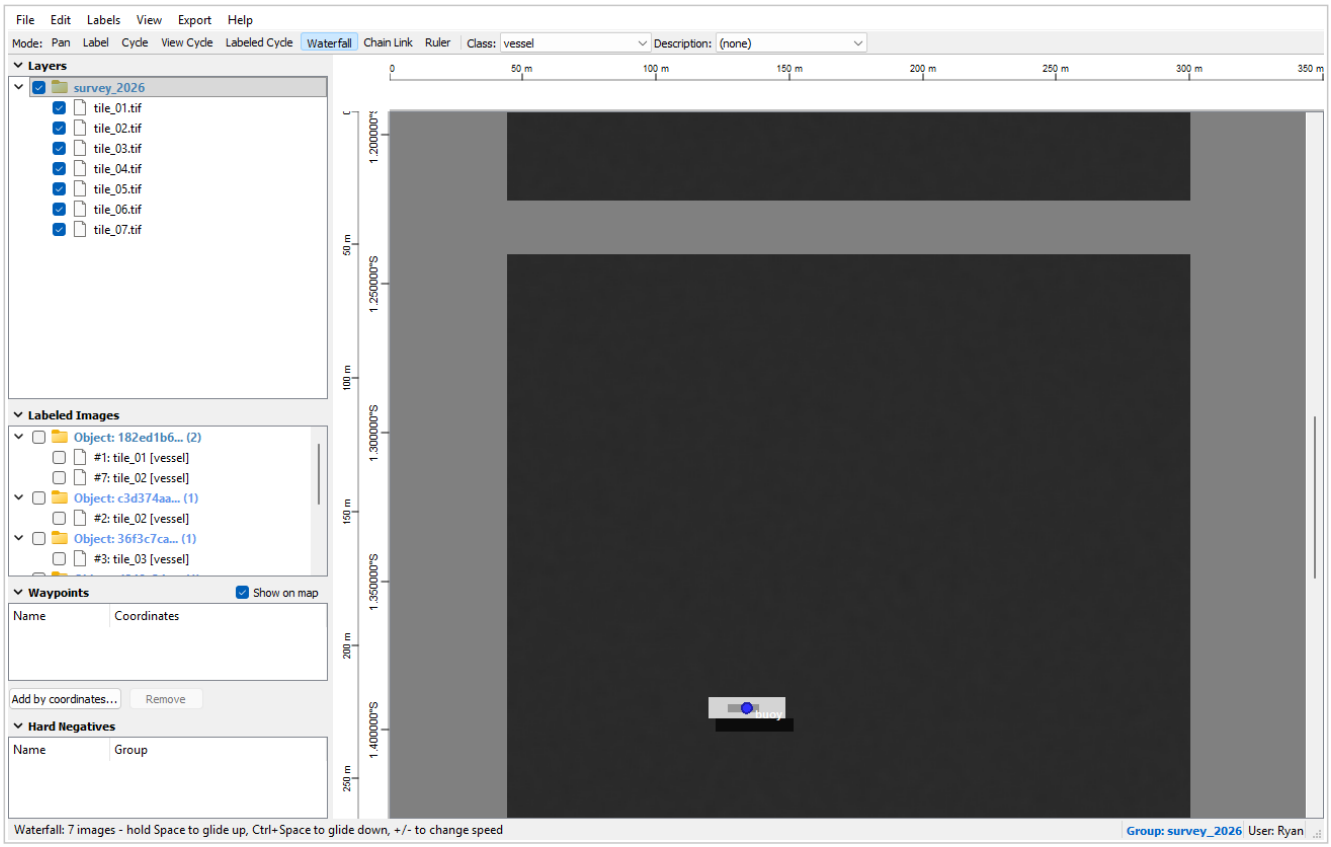
Stepping over the group's labelled tiles alone. Use it to check or correct work already done - after importing ground truth, or before an export - when the tiles with nothing on them would be wasted steps. Right-click > Move puts a label right without leaving the mode.



Labeled Cycle: only the tiles with labels are in the run; the status line counts them.

Waterfall (W): the group stacked as one strip

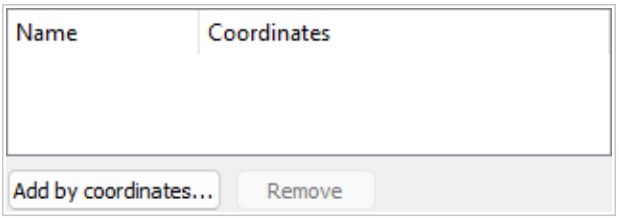
The group's images stacked vertically into a single strip, as a sonar or side-scan record is. Hold Space to glide up through it, Ctrl+Space to glide down; + and - change the speed. Use it when the imagery is one continuous track and an object may straddle two tiles: every label on every tile stays visible, and the boundaries between tiles stop mattering. Images load ahead of the glide.



Waterfall mode: the group's images stacked, labels on all of them visible.

Keeping your place; Go to Coordinates; Waypoints

Leaving a cycle keeps its place: come back to the mode and it resumes where it was, and the Labeled Images panel's checkboxes still work meanwhile. View > Go to Coordinates... (Ctrl+G) flies to a latitude and longitude. Right-click the map in Pan mode to add a Waypoint there - a named place to return to from the Waypoints panel, which Export > Waypoints (GPX)... writes for other tools.



The Waypoints panel.

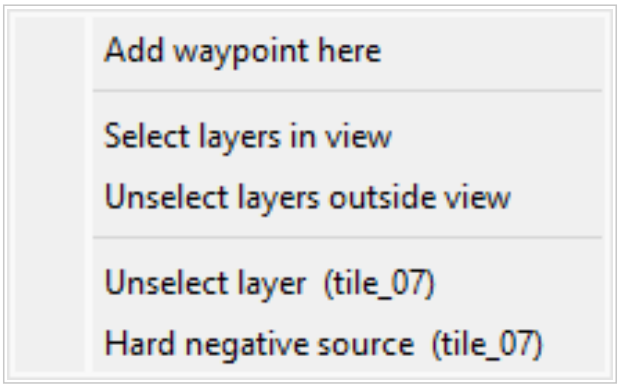
7 Flagging images with nothing in them: hard negatives

What the flag means, and what the HDF5 export does with it

A hard negative is an image that holds things a model might mistake for the object - rocks, wakes, debris - but no actual object. Flag such an image as a HARD NEGATIVE SOURCE and the HDF5 export can cut negative training examples from it: windows of the same size as the positive examples, slid across the image with a chosen overlap, each marked as not-an-object. Training on the things that look like the object but are not is what makes a model stop raising false alarms on them. An image with a label on it is never a source: no ground is taught as both an object and not one.

Flagging an image from the map

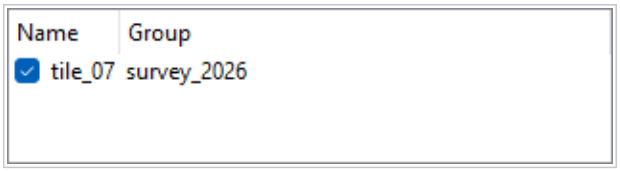
1. In Pan mode, right-click the image. The menu's last entry is 'Hard negative source (image name)': tick it.
2. Tick it again to take the flag off. Either way is one undo step.



The map's right-click menu, with the hard negative flag for the image under the cursor.

Reviewing the flagged set before an export

Every flagged image is listed in the Hard Negatives panel, so the whole set can be seen at a glance before it is used: the checkbox shows the image, a double-click zooms to it, and right-click > Remove hard negative flag takes one out. The flag is saved in the project. In the HDF5 export dialog, 'Negatives from' offers the flagged images as one of the choices (chapter 11).



The Hard Negatives panel with one image flagged.

8 Correcting a label: move it, reclass it, undo

Moving a label

A label in slightly the wrong place: right-click it > Move, then click where it belongs. On its own image only the position changes - masks, heading, size, rating and review stay. A click on another image asks first, because the label would change image and lose what was recorded on this one's pixels. The Snippet Editor's Move label tool does the same from the snippet (chapter 16).

Changing a label's class

Right-click > Change Class changes the whole linked object. In the Snippet Editor only this label can be changed if you prefer (chapter 22).

Undo and redo

Edit > Undo (Ctrl+Z) takes back the last edit, Redo (Ctrl+Y) puts it back; the menu names the step. The Snippet Editor's Ctrl+Z works on the same history.

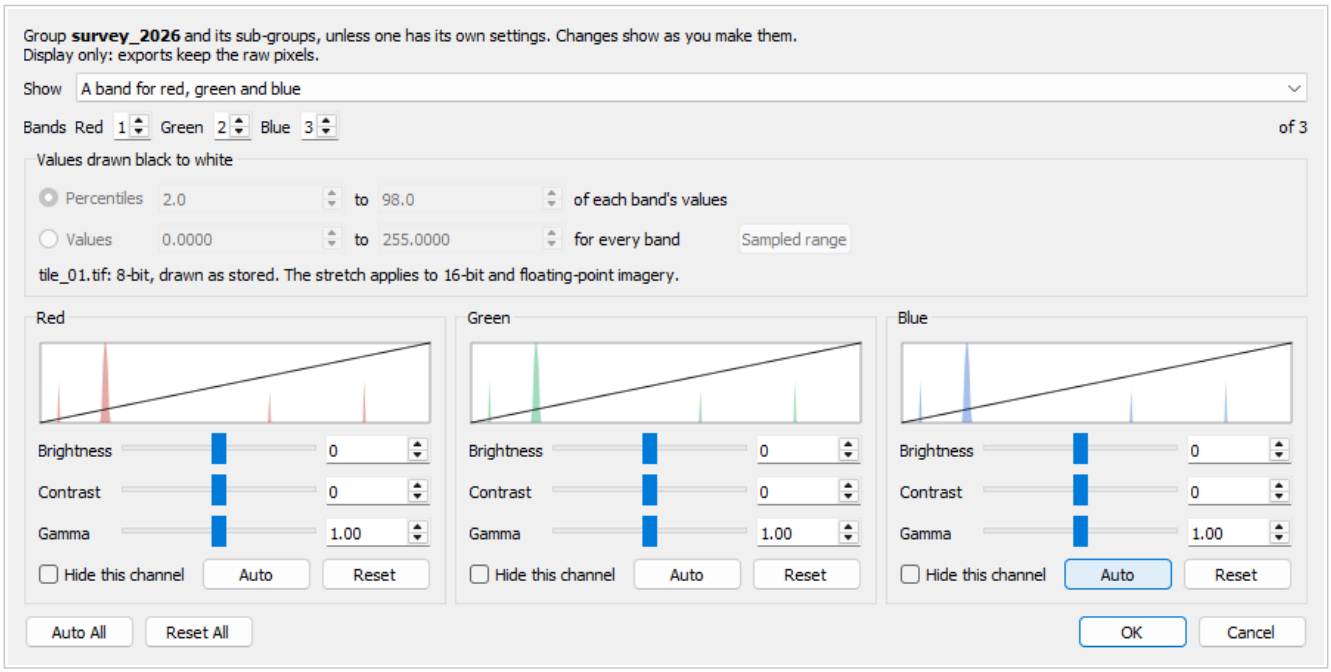


The status bar while a label is being moved, and the Edit menu naming the step Undo would take.

> Nearly everything is undoable: placing, removing, linking, moving and describing labels, masks (a step per stroke), headings, sizes, ratings, reviews, notes, class edits, hard negative flags, Clear All Labels, Import Ground Truth and Remove Missing Images. Adding imagery and display settings are not. Up to 200 steps are kept; opening another project clears them.

9 Choosing how the imagery is displayed

Multi-band imagery is drawn from the bands you choose, with brightness, contrast and gamma per channel, for the group selected in the Layers panel: View > Display Settings... (Ctrl+D), or right-click the group. A single band can be shown as greyscale. The settings are saved with the project and affect only what you see - exports and measurements read the imagery as it is. Imagery whose file names a colour for each pixel value - a palette, as a classified raster or an indexed-colour image carries - is drawn in those colours by default, as QGIS draws it; the window's COLOUR TABLE tick turns that off for the group, and the values are drawn as grey instead. Its LEGEND row lists the values the imagery holds, each with its colour and its share of the pixels.



View > Display Settings... for a group.

16-bit and floating-point imagery: what is drawn black and white

8-bit imagery is drawn as stored. Anything else is stretched for display: by default each band's values between its 2nd and 98th percentiles run from black to white, and the window lets you move those percentiles. Imagery that is already a fraction - values from 0 to 1 - is better drawn as it is: choose Values 0 to 1, which is simply times 255. The window says what the imagery's values look like, suggests 0 to 1 when they lie within it, and Sampled range fills in the lowest and highest values found. For 8-bit imagery these controls are greyed out.

10 Saving the project, and imagery that has moved

Saving, and the recovery copy

File > Save Project (Ctrl+S) writes the project file; Save Project As... chooses the name. The title bar shows a * while there is unsaved work. A recovery copy is written as you work, and offered back after a crash.

Imagery that has moved, or is not coming back

A project refers to imagery by path. Opened on another machine, or after the imagery was moved, File > Locate Missing Images... points it at the new folder. File > Remove Missing Images... takes out the images that are not coming back, with their labels (undoable).

Imagery cropped since it was labelled

A copy of the imagery with columns or rows cut off - the same files by name, smaller, their georeferencing updated to match - puts every label a fixed number of pixels off. When a project opens on such files, GeoLabeller notices the size difference and offers to re-seat the labels; accepting also brings every other image's record up to date in the background, nothing to move and nothing asked, so a project saved afterwards matches the cropped files throughout. By the georeferencing the crop is a whole-pixel shift, and every label and mask moves by it; labels in the cut-off part are dropped, and the project is told the new size. If the pixel size changed as well, the labels are placed by their latitude and longitude after a question, and masks, painted on the old pixels, are dropped. One undo step. File > Re-seat Labels to Imagery... does it any time, and Locate Missing Images accepts a cropped copy of an image as that image.

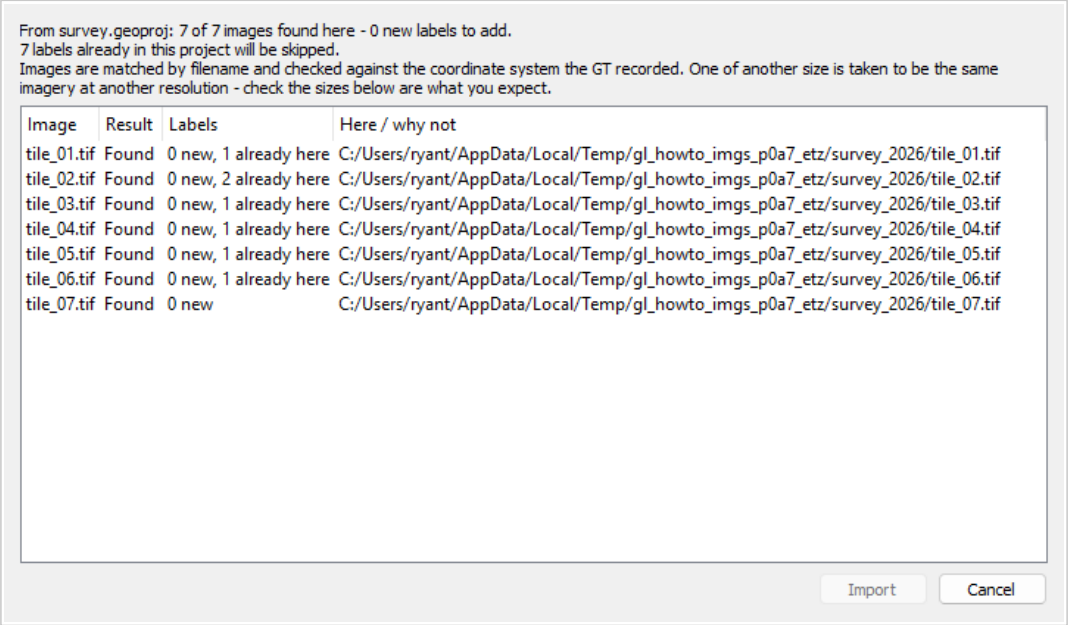
Sharing a machine, and combining projects

File > Switch User... sets the name recorded on everything you do, so a shared machine still says who did what. File > Combine Projects... merges another project into this one.

11 Bringing ground truth in and sending it out

Importing labels made elsewhere

File > Import Ground Truth... adds the labels from another project or ground-truth file onto the images loaded here, matched by filename. The dialog previews what will happen before anything changes: which images were found, which are missing, and how many labels each brings.

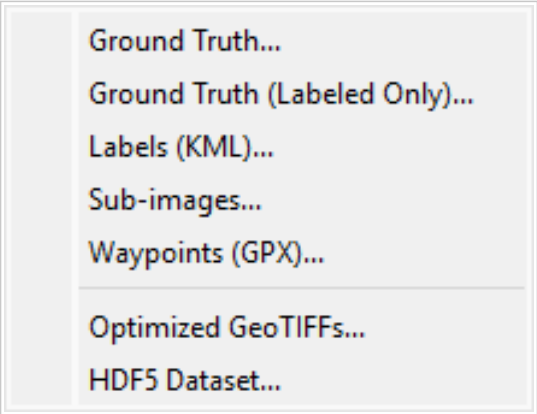


The Import Ground Truth preview.

A label brought in may land on an object already labelled here - the same image, within a few pixels. File > Import Options... says what then happens, for Import Ground Truth, Combine Projects and a Model Review import alike: keep the labels already here and drop the new (the default), keep the new and drop the old, keep both, or ask - a window lists each pair for a decision by hand. The neighbourhood that counts as the same object is a setting too, in pixels of the image. What is dropped goes with the import's undo step.

> Ground truth made on downsampled copies of the imagery can be imported onto the full-resolution originals: an image of the same name and another size is taken as the same imagery at another resolution, and each label is placed at the same ground position. Masks are left out where sizes differ. Then Labeled Cycle (Shift+C) and Move are the way to check and correct any that are off.

Exporting ground truth and sub-images

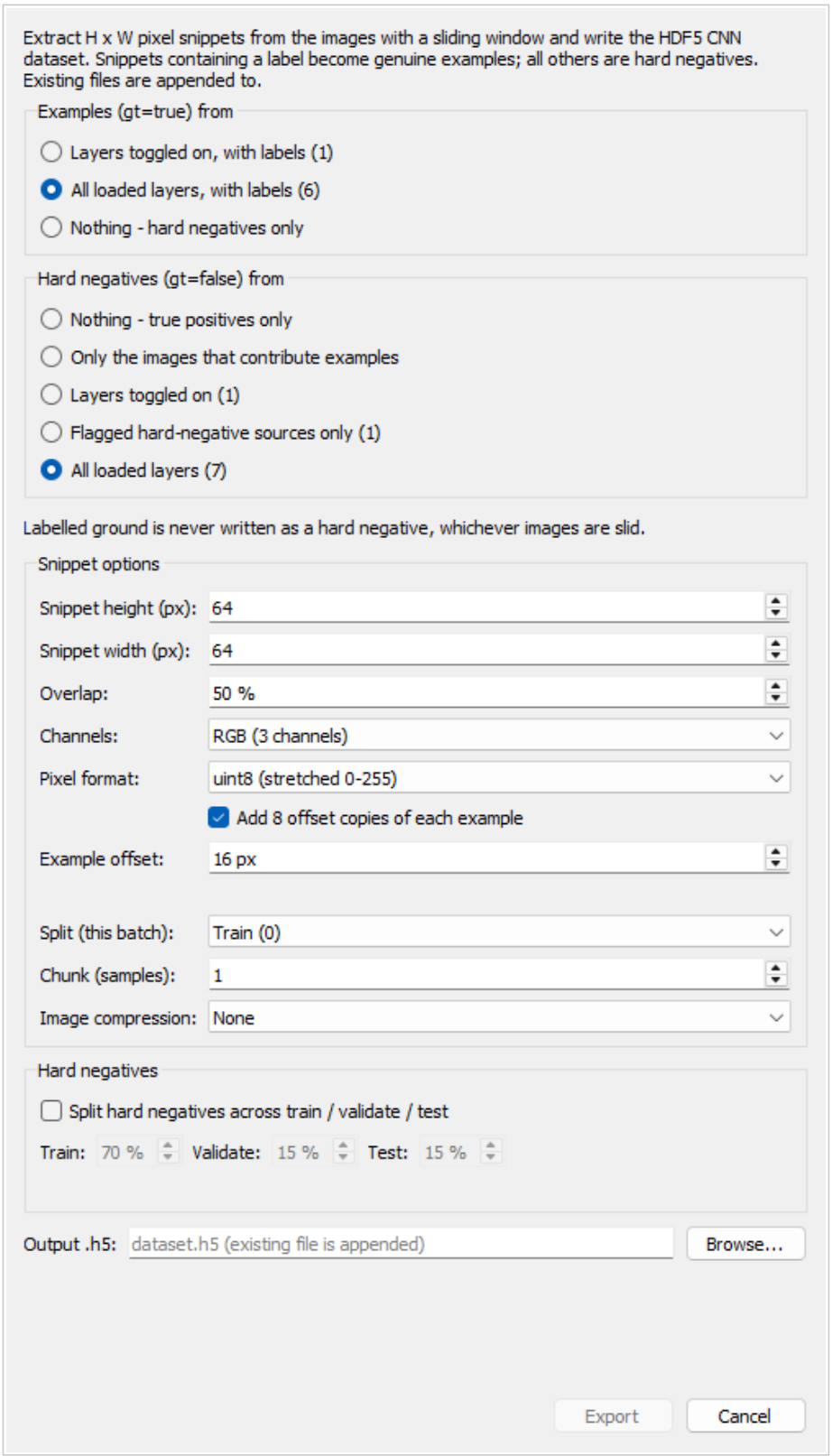


The Export menu.

- Export > Ground Truth... writes every label (positions, classes, links, headings, sizes, masks, ratings, reviews) to a file another GeoLabeller - or your own tools - can read. Ground Truth (Labeled Only) leaves out images with no labels.
- Export > Sub-images... cuts a GeoTIFF round every label at a ground size you give.
- Export > Optimized GeoTIFFs... writes tiled, pyramided copies of the imagery that load far faster next time.
- Export > Waypoints (GPX)... for GPS units and GIS tools.
- Export > Labels (KML)... writes every label that has a position on the ground - a folder per class in its colour, each label with its image, pixel position, size, heading, rating and who made it - for Google Earth or a GIS. Rejected labels stay out, as they do from the HDF5 dataset.

Exporting the HDF5 training dataset

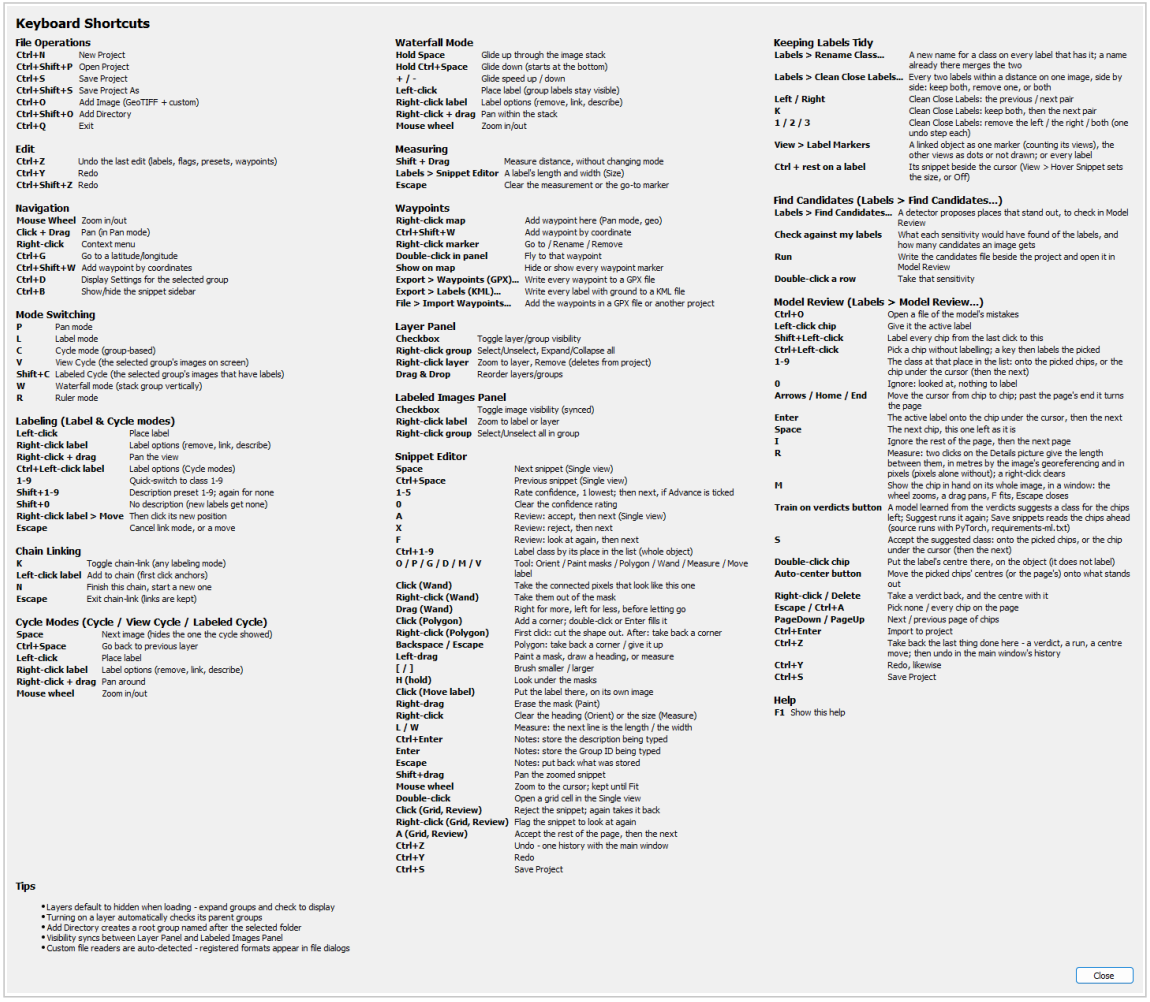
Export > HDF5 Dataset... writes the training dataset: a chip per label at a chosen size, with its class, masks, heading, size, rating and provenance; and negatives, cut from the images you choose - the labelled ones, the visible ones, every image, or only the flagged hard negative sources (chapter 7). Rejected labels are left out. The dialog estimates what the settings will produce before it writes, and a later export can append to the same file with a different split.



Export > HDF5 Dataset...

12 Every key on the map

Help > Keyboard Shortcuts (F1) shows every key. The tables below are taken from the same source, so they match what F1 shows.



Help > Keyboard Shortcuts (F1).

File Operations

Key	Does
Ctrl+N	New Project
Ctrl+Shift+P	Open Project
Ctrl+S	Save Project
Ctrl+Shift+S	Save Project As
Ctrl+O	Add Image (GeoTIFF + custom)
Ctrl+Shift+O	Add Directory
Ctrl+Q	Exit

Edit

Key	Does
Ctrl+Z	Undo the last edit (labels, flags, presets, waypoints)
Ctrl+Y	Redo
Ctrl+Shift+Z	Redo

Navigation

Key	Does
Mouse Wheel	Zoom in/out
Click + Drag	Pan (in Pan mode)
Right-click	Context menu
Ctrl+G	Go to a latitude/longitude
Ctrl+Shift+W	Add waypoint by coordinates
Ctrl+D	Display Settings for the selected group
Ctrl+B	Show/hide the snippet sidebar

Mode Switching

Key	Does
P	Pan mode
L	Label mode
C	Cycle mode (group-based)
V	View Cycle (the selected group's images on screen)
Shift+C	Labeled Cycle (the selected group's images that have labels)
W	Waterfall mode (stack group vertically)
R	Ruler mode

Labeling (Label & Cycle modes)

Key	Does
Left-click	Place label
Right-click label	Label options (remove, link, describe)
Right-click + drag	Pan the view
Ctrl+Left-click label	Label options (Cycle modes)
1-9	Quick-switch to class 1-9
Shift+1-9	Description preset 1-9; again for none
Shift+0	No description (new labels get none)
Right-click label > Move	Then click its new position
Escape	Cancel link mode, or a move

Chain Linking

Key	Does
K	Toggle chain-link (any labeling mode)
Left-click label	Add to chain (first click anchors)
N	Finish this chain, start a new one
Escape	Exit chain-link (links are kept)

Cycle Modes (Cycle / View Cycle / Labeled Cycle)

Key	Does
Space	Next image (hides the one the cycle showed)
Ctrl+Space	Go back to previous layer

Left-click	Place label
Right-click label	Label options (remove, link, describe)
Right-click + drag	Pan around
Mouse wheel	Zoom in/out

Waterfall Mode

Key	Does
Hold Space	Glide up through the image stack
Hold Ctrl+Space	Glide down (starts at the bottom)
+ / -	Glide speed up / down
Left-click	Place label (group labels stay visible)
Right-click label	Label options (remove, link, describe)
Right-click + drag	Pan within the stack
Mouse wheel	Zoom in/out

Measuring

Key	Does
Shift + Drag	Measure distance, without changing mode
Labels > Snippet Editor	A label's length and width (Size)
Escape	Clear the measurement or the go-to marker

Waypoints

Key	Does
Right-click map	Add waypoint here (Pan mode, geo)
Ctrl+Shift+W	Add waypoint by coordinate
Right-click marker	Go to / Rename / Remove
Double-click in panel	Fly to that waypoint
Show on map	Hide or show every waypoint marker
Export > Waypoints (GPX)...	Write every waypoint to a GPX file
Export > Labels (KML)...	Write every label with ground to a KML file
File > Import Waypoints...	Add the waypoints in a GPX file or another project

Layer Panel

Key	Does
Checkbox	Toggle layer/group visibility
Right-click group	Select/Unselect, Expand/Collapse all
Right-click layer	Zoom to layer, Remove (deletes from project)
Drag & Drop	Reorder layers/groups

Labeled Images Panel

Key	Does
Checkbox	Toggle image visibility (synced)
Right-click label	Zoom to label or layer

Right-click group	Select/Unselect all in group

Keeping Labels Tidy

Key	Does
Labels > Rename Class...	A new name for a class on every label that has it; a name already there merges the two
Labels > Clean Close Labels...	Every two labels within a distance on one image, side by side: keep both, remove one, or both
Left / Right	Clean Close Labels: the previous / next pair
K	Clean Close Labels: keep both, then the next pair
1 / 2 / 3	Clean Close Labels: remove the left / the right / both (one undo step each)
View > Label Markers	A linked object as one marker (counting its views), the other views as dots or not drawn; or every label
Ctrl + rest on a label	Its snippet beside the cursor (View > Hover Snippet sets the size, or Off)

Find Candidates (Labels > Find Candidates...)

Key	Does
Labels > Find Candidates...	A detector proposes places that stand out, to check in Model Review
Check against my labels	What each sensitivity would have found of the labels, and how many candidates an image gets
Run	Write the candidates file beside the project and open it in Model Review
Double-click a row	Take that sensitivity

Help

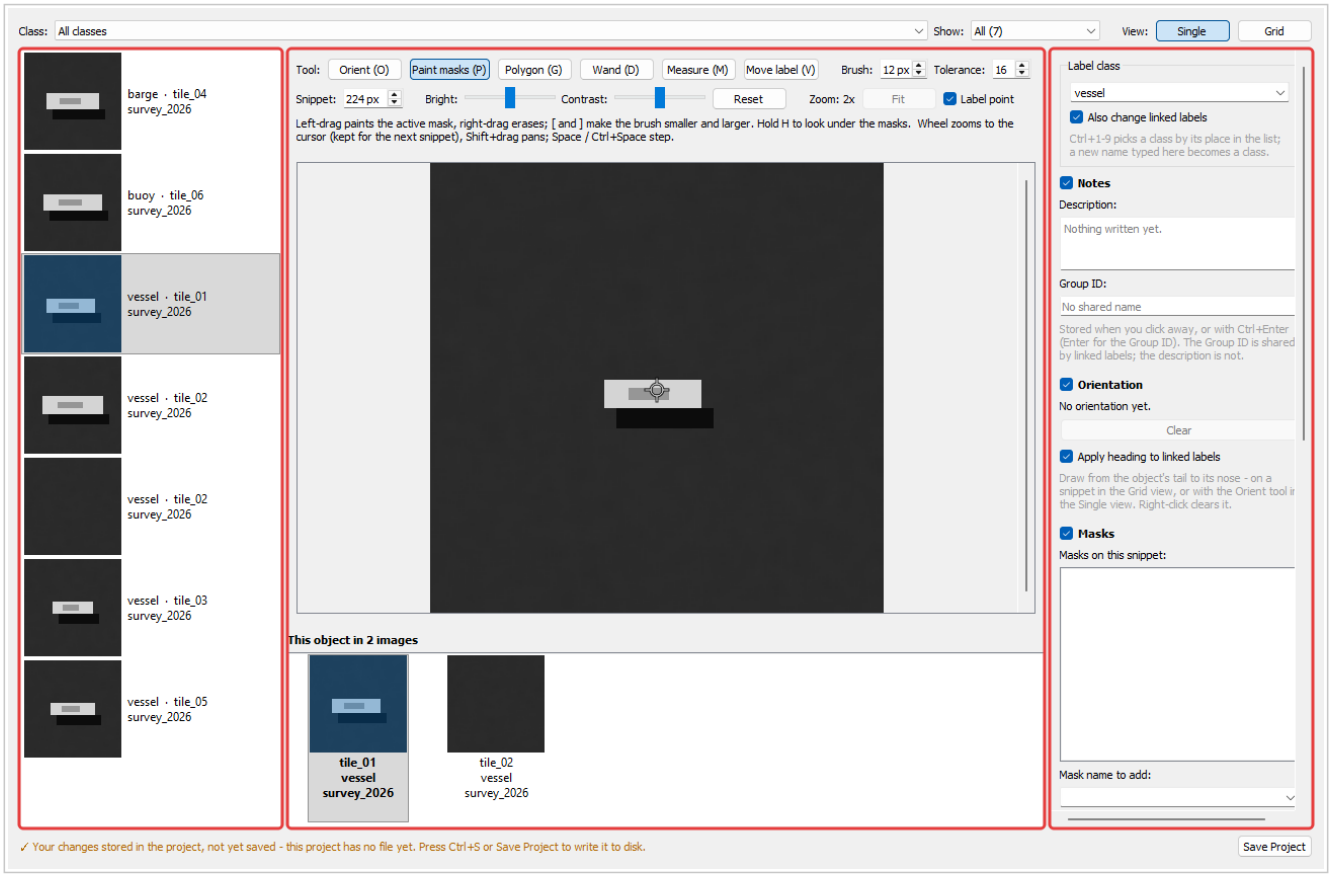
Key	Does
F1	Show this help

Part 2

The Snippet Editor

13 What the Snippet Editor is for, and a tour of its window

A label on the map is a point and a class. The Snippet Editor is where everything else about the object is recorded, one snippet at a time: its outline as masks, its heading, its length and width, how sure you are of it, whether it has been reviewed, and any notes. Open it with Labels > Snippet Editor.... It shows the labels of the open project; what you do in it is stored in the project as you go, and the bar at the bottom says whether that is on disk yet.



The Snippet Editor. Left: the list of snippets. Middle: the Single view - one snippet, zoomable, with the tools. Right: the column of sections, one per thing recorded.

The column of sections, and switching them off

Each section on the right switches on and off by its header. Switching one off takes all of it away - its panel, its entry on the Show filter, its overlay on the snippets, its tool and its keys - so the editor shows only what your project records.

14 Choosing which snippets to see, and stepping through them



Class chooses which labels are listed; Show narrows them to what still needs doing; View switches Single and Grid.

The Class and Show filters

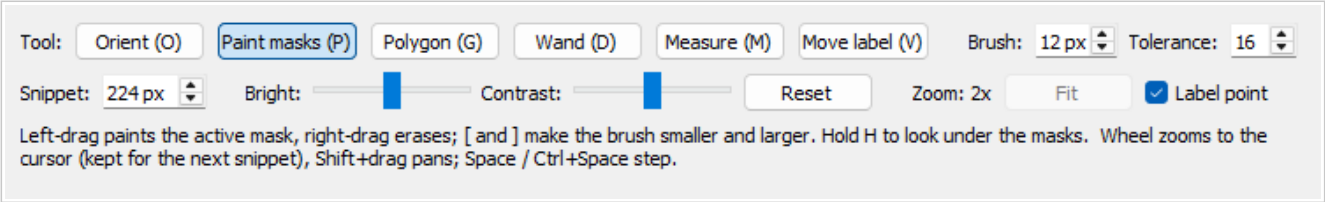
- CLASS lists one class, all classes, or one linked object (every object is offered after the classes, named by its class, group ID and number of views).
- SHOW offers, for every section that is on, the snippets that still need it and the ones that have it: Needs masks / Masked, Needs orientation / Oriented, and so on, with live counts. A finished snippet leaves a Needs list when you move off it, so the one in hand never vanishes under you.

Stepping with Space

Space steps to the next snippet and Ctrl+Space to the previous, wherever the keyboard focus is. Clicking a row works too. Thumbnails are read as they come into view, so a project of thousands of labels opens at once.

15 Outlining the object with masks

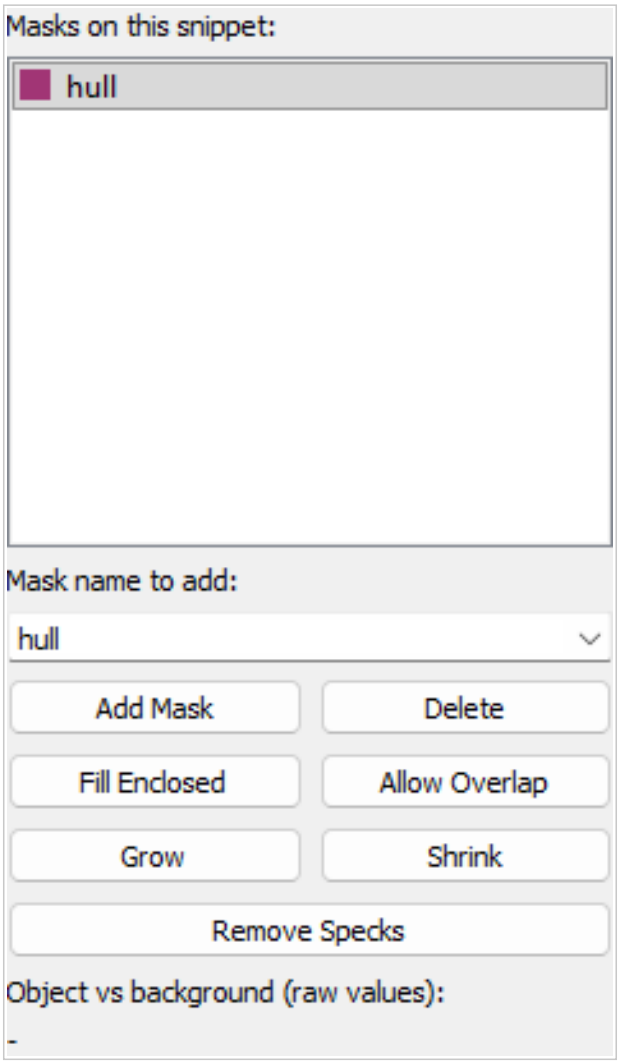
A mask is a named binary layer painted over the snippet - 'hull', 'shadow' - stored in the project anchored to the imagery, so it stays put whatever snippet size you look at it in. Several masks can sit on one snippet; by default they do not overlap, so other masks act as walls for whatever you draw.



The tool row (one button per tool, its key in brackets), the view row, and the line that says what the tool in hand does.

Adding a named mask

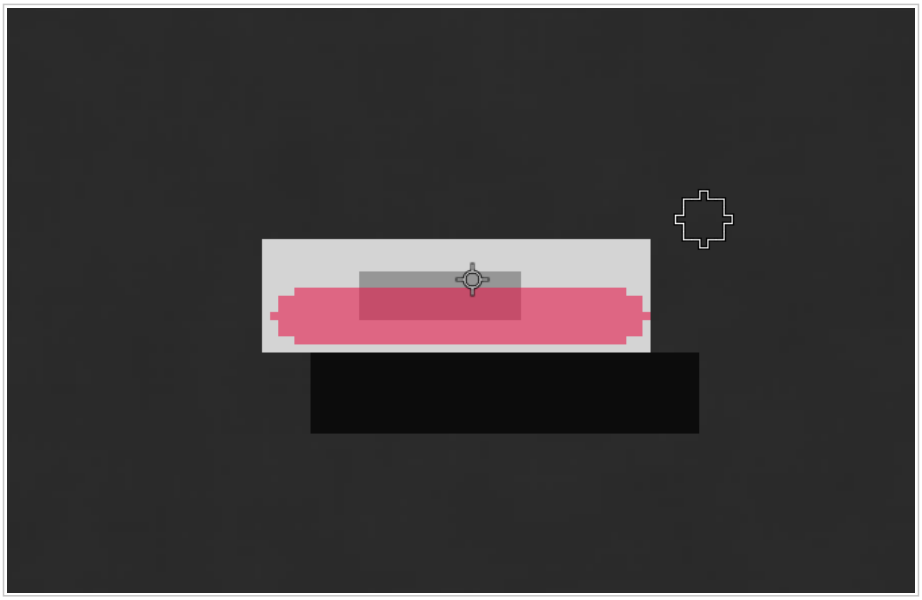
1. In the Masks section, pick or type the name under 'Mask name to add' and press Add Mask. The name joins the project's presets (Labels > Edit Mask Names...), so the next snippet offers it.
2. The new mask is the active one - the one the tools paint. Choosing a name the snippet already has selects that mask.



The Masks section.

Painting with the brush: Paint masks (P)

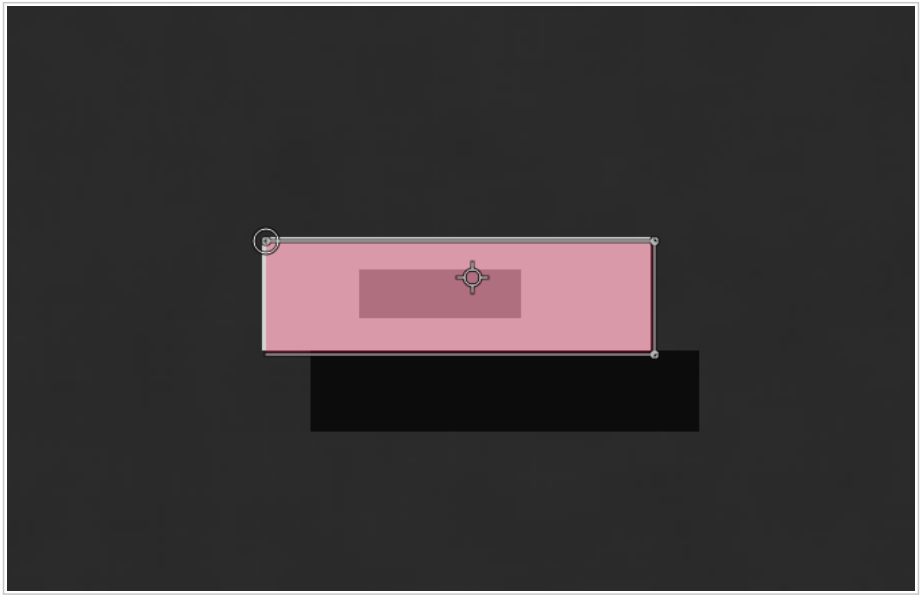
Left-drag paints the active mask, right-drag erases. The outline under the cursor shows exactly which pixels the next stamp will paint. [and] make the brush smaller and larger.



A brush stroke, and the brush outline.

Clicking a shape out: Polygon (G)

1. Click round the object, corner by corner. The shape fills in as you go.
2. Double-click, press Enter, or click the first corner to finish: the shape is filled into the active mask.
3. Right-click or Backspace takes back the last corner; Escape gives the shape up.
4. To cut a shape OUT of the mask, begin it with a right-click.

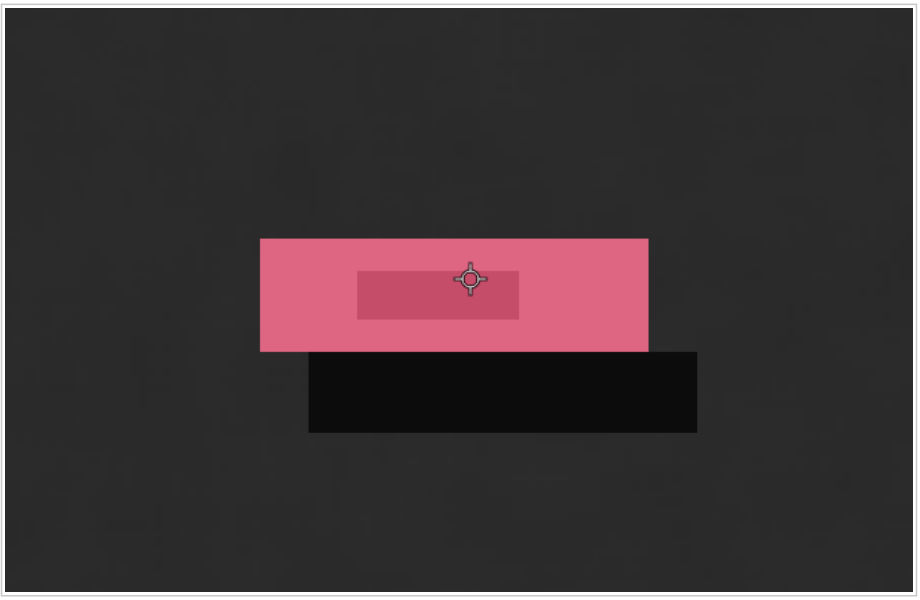


A polygon being clicked out: three corners placed, the fourth following the cursor, the ring marking where a click closes it.

Taking a patch of like pixels: Wand (D)

1. Click the object. The connected pixels that look like the one you clicked light up - then join the active mask when you let go. A right-click takes them out of the mask instead.

2. Too little or too much? Keep the button down and drag right to take in more, left to take in less. The Tolerance box follows, and the next click starts from where you ended.

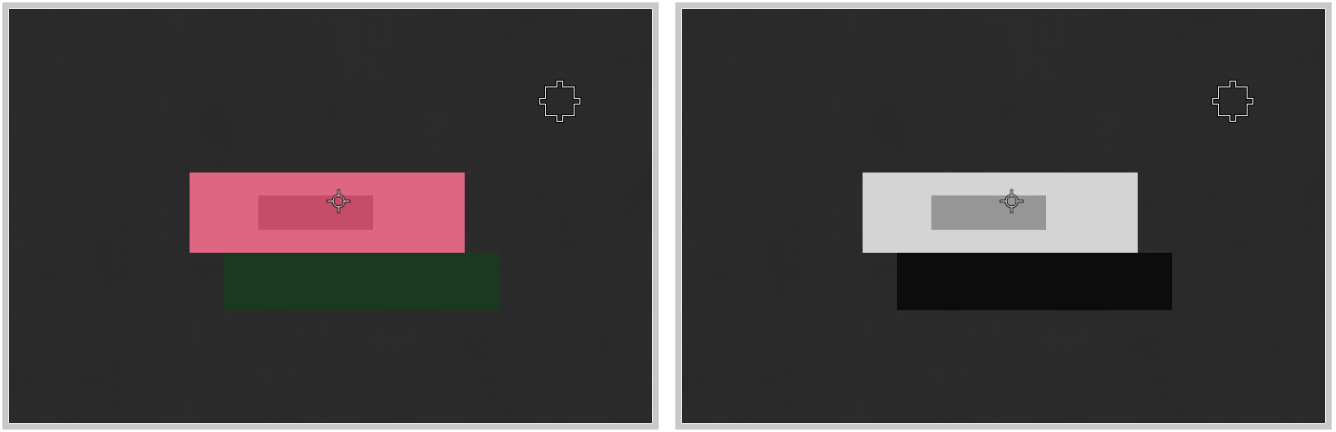


The wand held down and dragged right: the hull and its wheelhouse are both within tolerance now and shown as what will be taken.

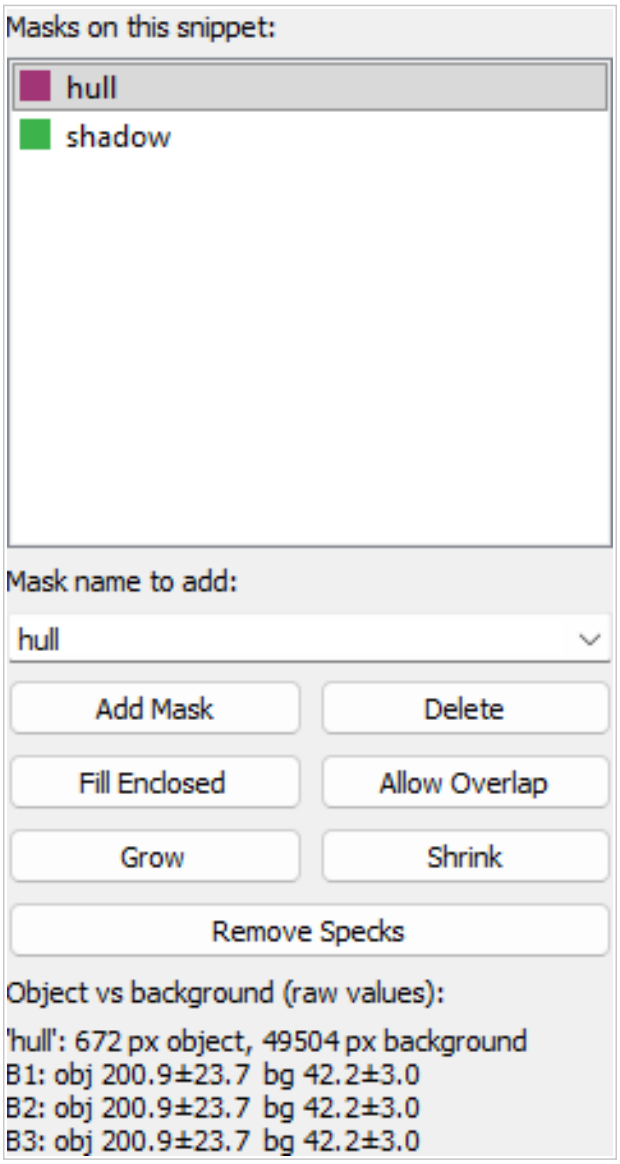
> 'Looks like' is judged on the snippet as you see it, brightness and contrast included. Raise the contrast and the wand tells apart what it otherwise would not.

Tidying: Grow, Shrink, Remove Specks, Fill Enclosed; looking under the masks (H)

- GROW and SHRINK make the active mask one pixel wider or narrower all round.
- REMOVE SPECKS drops the separate pieces smaller than the brush - set the brush size to say how small. The largest piece always stays, and the button says what it found.
- FILL ENCLOSED fills the holes: the inside of a drawn outline, or what the wand left out. It refuses when the outline has a gap.
- Grow, Fill Enclosed, Shrink closes a ragged mask in three clicks.
- ALLOW OVERLAP lets masks share pixels; off, other masks are walls for everything you draw, and edges for Fill Enclosed.
- Hold H to look at the imagery under the masks.



The same snippet with its two masks drawn, and with H held.



Two masks on a snippet - hull and shadow - and the tidying buttons. Below them, the object's raw pixel values against the background.

> Every stroke, shape, wand click and tidy is one undo step, and is stored in the project the moment it is made.

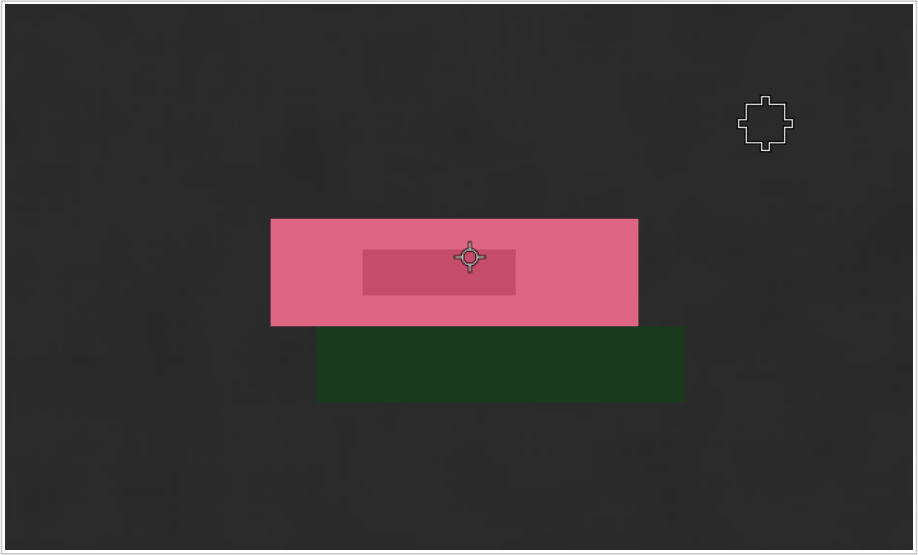
16 Looking closely: zoom that stays, the label point, moving the label



The view row: the snippet size, brightness and contrast (display only), the zoom with Fit, and the Label point.

Zoom, Fit, snippet size, brightness and contrast

- The WHEEL zooms to the cursor, 0.25x to 32x; Shift+drag pans. The zoom you set stays for the next snippet, with the view held on the label, so a pass of fine work needs no re-zooming. FIT goes back to the whole snippet.
- SNIPPET is how much of the image is shown round the label, in source pixels. Masks are anchored to the imagery, so changing it only changes the window you work in.
- BRIGHT and CONTRAST change the display only; Reset puts them back.



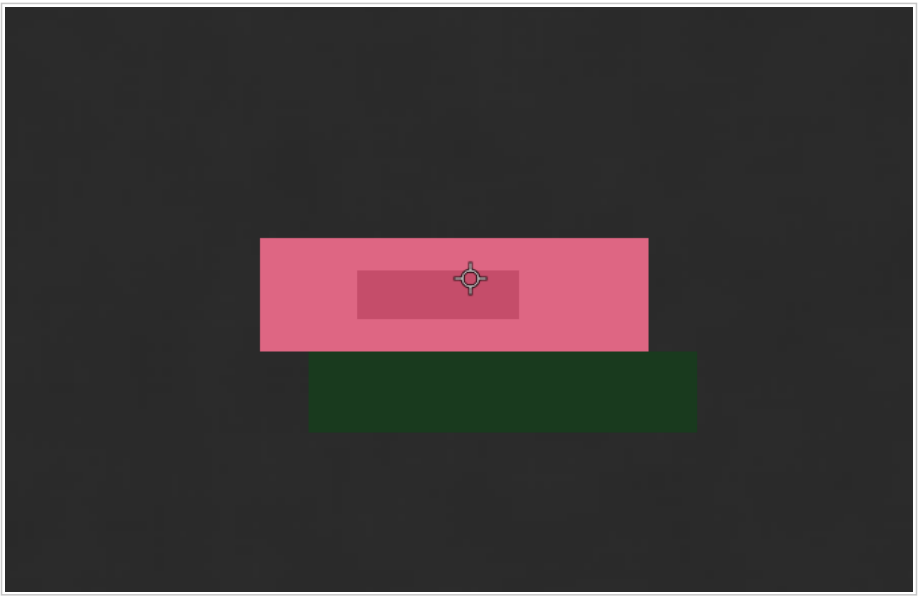
Zoomed in on the hull.

The label point

LABEL POINT marks where the label itself is - a ring, open in the middle. A snippet is cut round its label, so the point is usually the middle; near an image's edge it is not, and the ring says so.

Moving the label from the snippet: Move label (V)

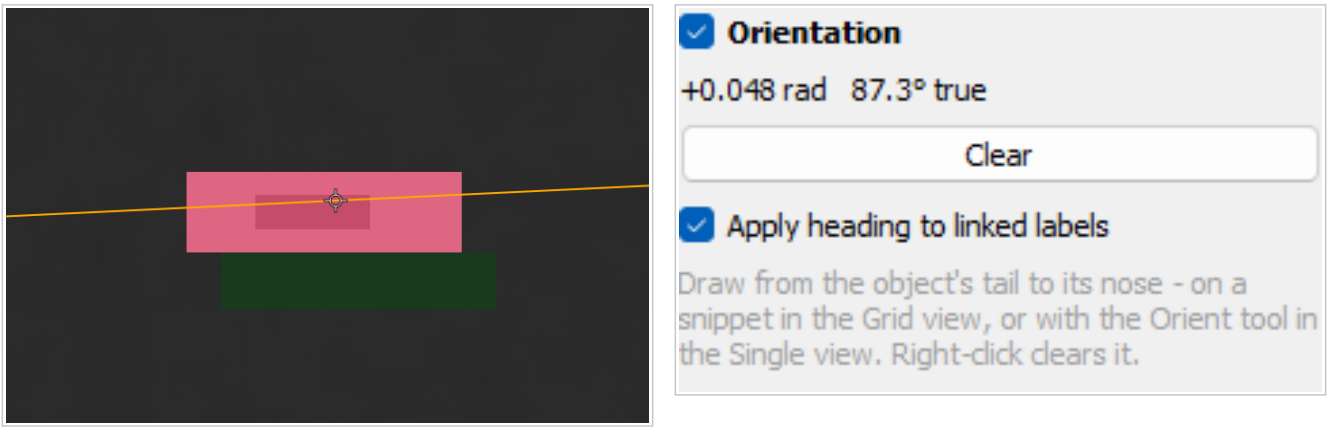
Click where the label belongs and it moves there, on its own image, to a fraction of a pixel when zoomed. Masks, heading, size, rating and review stay as they were. The tool stays in hand, so correcting a run of positions is Space, click, Space, click.



The Move label tool: the ring is the label's own point; a click puts it elsewhere.

17 Recording the object's orientation (Orient, O)

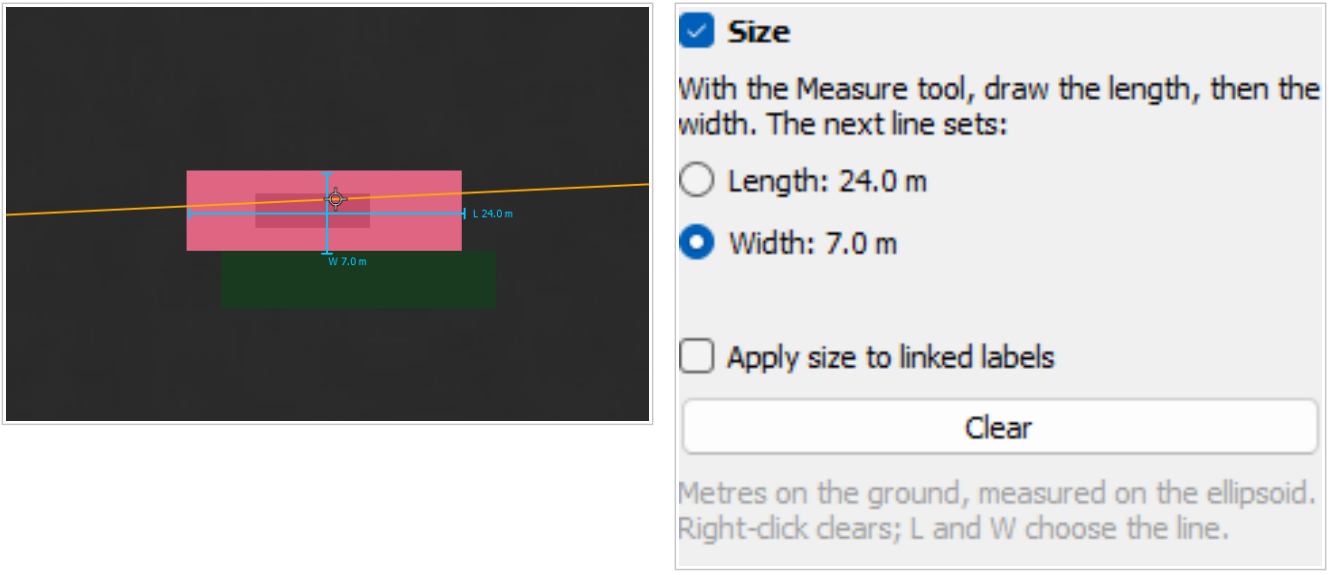
With the Orient tool, drag across the snippet from the object's tail to its nose. The pixel angle is stored, and for georeferenced imagery the true-north heading too. Right-click clears it. The Orientation section reads out the heading, and with 'Apply heading to linked labels' ticked the other views of a linked object take the same true-north heading, each worked out through its own image (shown violet until drawn over by hand). The Grid view orients a page at a time (chapter 23).



An orientation drawn with the Orient tool, and the Orientation section.

18 Measuring the object's size: length and width (Measure, M)

With the Measure tool, draw the length across the object, then the width: the first line is the length, the next the width, in metres on the ground. L and W choose which the next line sets; right-click clears both. The Size section shows them and can share them with the linked labels.



A length and a width measured, and the Size section.

19 Rating how sure you are (Confidence, 1 to 5)

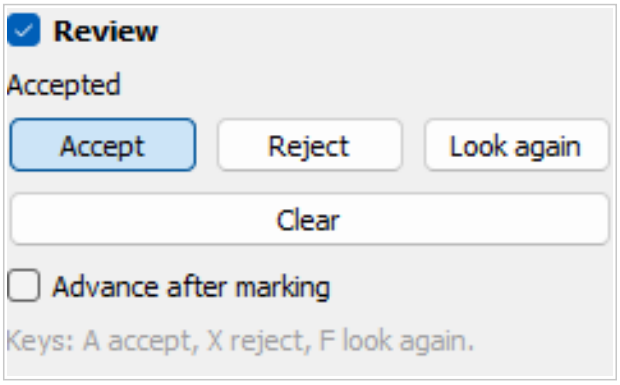
How sure you are of the label, 1 (lowest) to 5 (highest): the buttons, or keys 1 to 5. 0 clears it. Unrated labels export as 0. Tick 'Advance after rating' and a rating key moves on to the next snippet, so a rating pass is one key a snippet.



The Confidence section.

20 Reviewing labels: accept, reject, look again

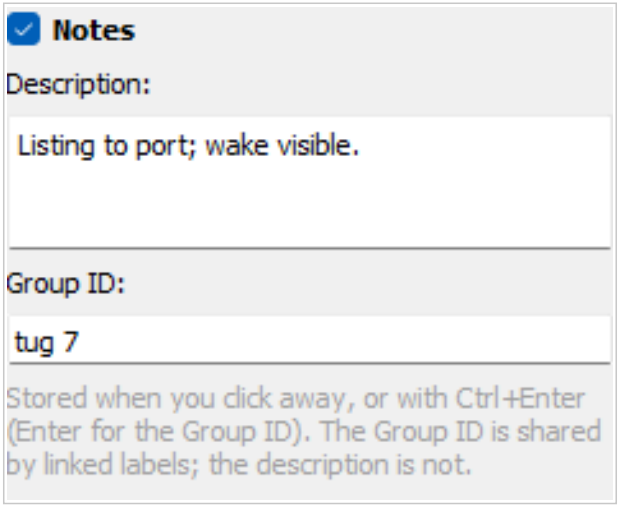
Whether somebody has checked the label, and what they found: Accept (A), Reject (X) or Look again (F). With 'Advance after marking' on, a key also moves to the next snippet. A rejected label stays in the project and the ground truth, is drawn dimmed on the map, and is left out of the HDF5 and sub-image exports. For reviewing many at once, see the Grid view's Review mode (chapter 23).



The Review section.

21 Writing notes: the description and the Group ID

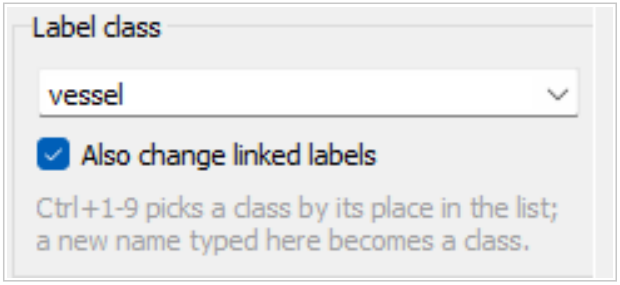
The label's description and its object's Group ID, shown and edited in place. The description is this label's alone; the Group ID is one name for every label linked to it. A note is stored when you click away, or with Ctrl+Enter (Enter for the Group ID); Escape puts back what was stored.



The Notes section.

22 Changing a snippet's class

The Label class box gives the snippet in hand another class; Ctrl+1 to Ctrl+9 pick by place in the class list. A new name typed here becomes a class. With 'Also change linked labels' ticked the whole linked object changes; unticked, only this snippet does - for the part of an object that needs a label of its own.



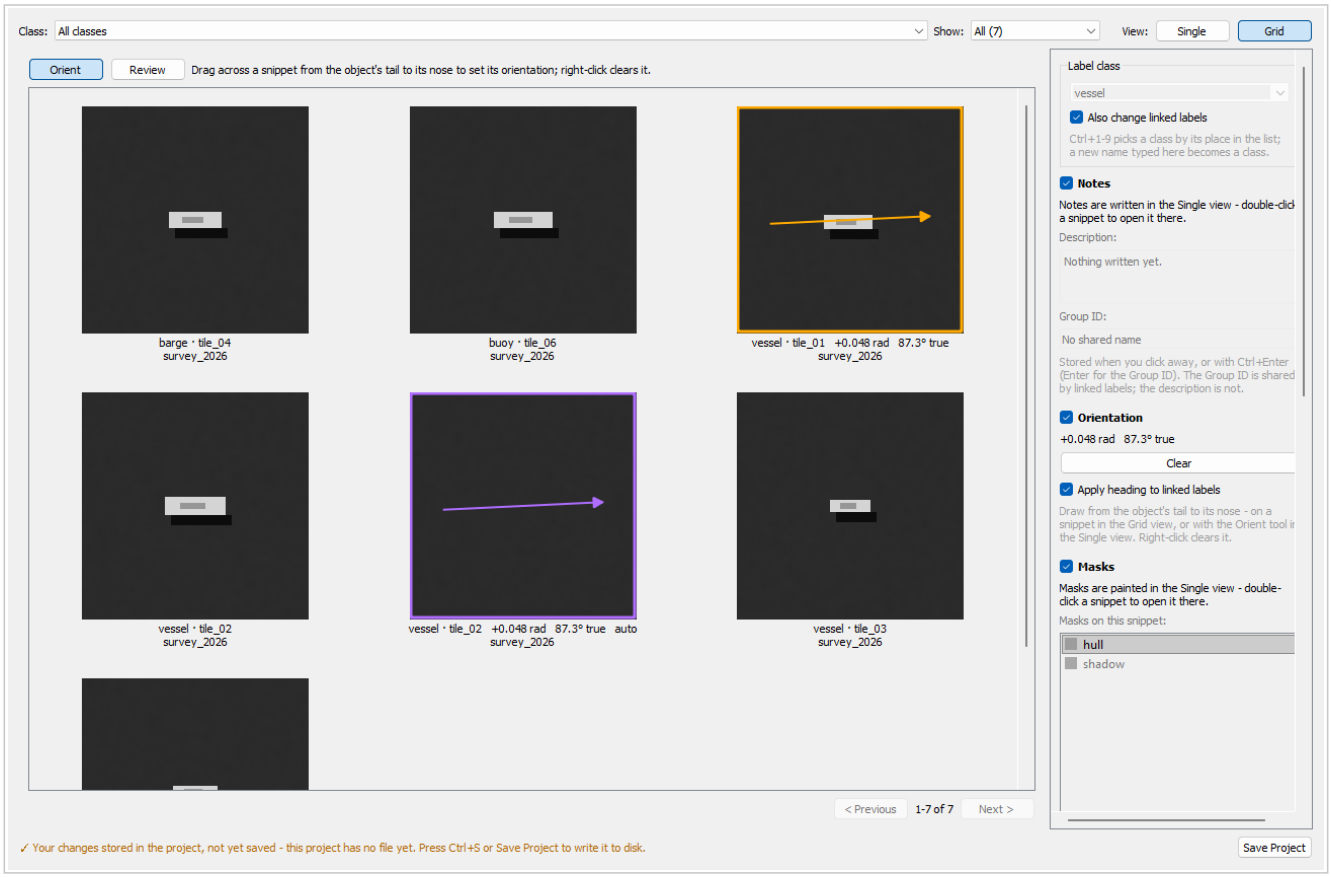
The Label class box.

23 Working a page at a time: the Grid view

The Grid view lays out a page of sixty snippets at once, in the list's order, and has two modes. Double-click any snippet to open it in the Single view.

Orient mode: a page of headings

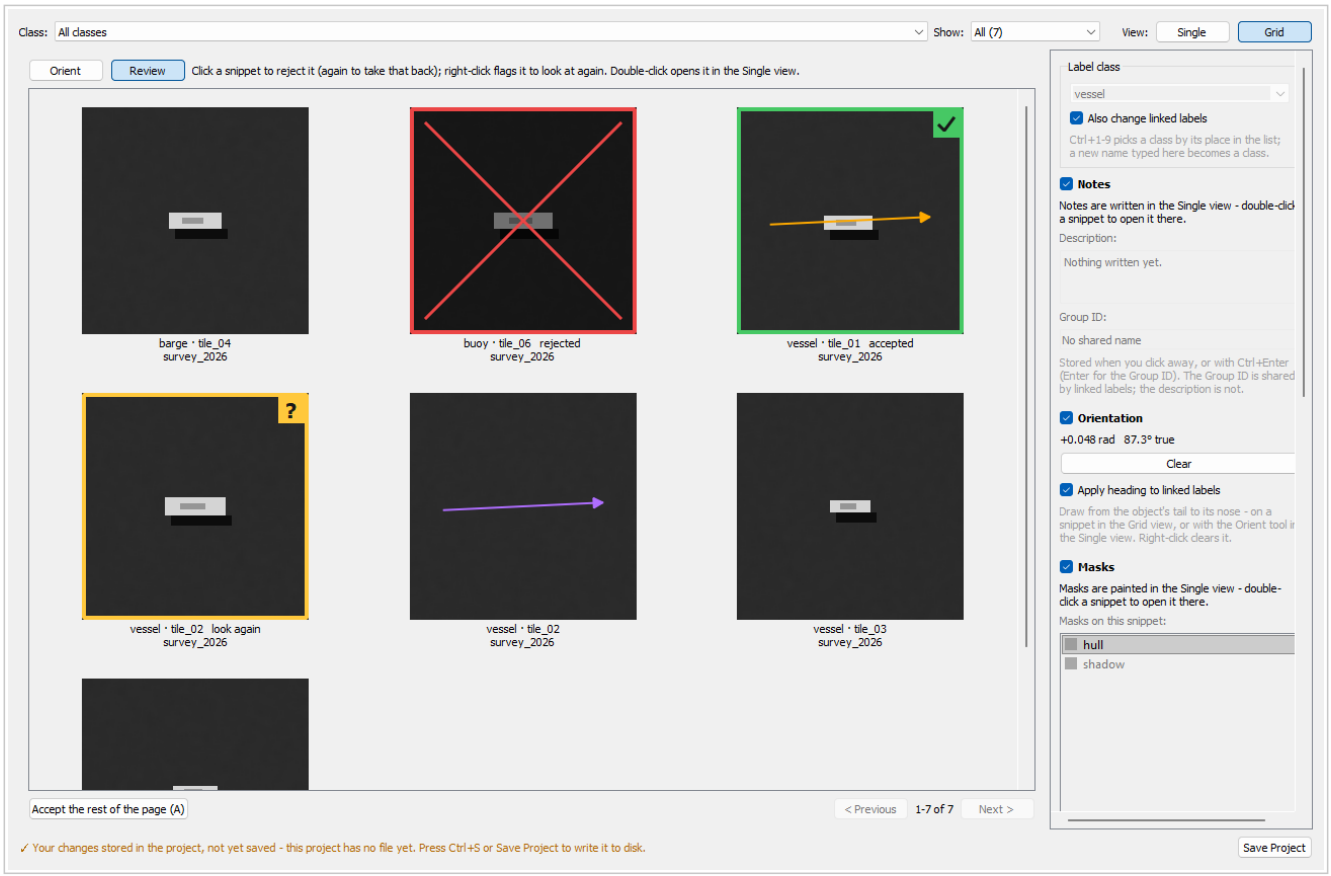
Drag across a snippet to set its heading, as with the Orient tool - a whole page oriented without opening any. Oriented snippets wear a border: amber drawn by hand, violet given by a linked label.



The Grid view in Orient mode.

Review mode: a page of reviews

Each snippet wears what its review found. Click one to reject it (again to take that back), right-click to flag it to look at again, and press A - or 'Accept the rest of the page' - to accept everything left unmarked and go on to the next page. Accepting a page is one undo step.



The Grid view in Review mode: one snippet rejected, one flagged, the rest ready for 'Accept the rest of the page'.

24 Seeing every view of one object: the linked row

When the snippet in hand belongs to a linked object, a row under the Single view shows every view of that object, captioned with its image and what has been recorded on it. Click one to put it in hand. Choosing the object in the Class box lists only its views.



The linked row under the Single view.

25 Saving, and every key in the editor

Everything you do in the editor is stored in the project at once; the bar at the bottom says whether the project is on disk yet. Save Project (Ctrl+S) writes it, from either window.

The save bar.

Every key the editor takes

Key	Does
Space	Next snippet (Single view)
Ctrl+Space	Previous snippet (Single view)
1-5	Rate confidence, 1 lowest; then next, if Advance is ticked
0	Clear the confidence rating
A	Review: accept, then next (Single view)
X	Review: reject, then next
F	Review: look at again, then next
Ctrl+1-9	Label class by its place in the list (whole object)
O / P / G / D / M / V	Tool: Orient / Paint masks / Polygon / Wand / Measure / Move label
Click (Wand)	Take the connected pixels that look like this one
Right-click (Wand)	Take them out of the mask
Drag (Wand)	Right for more, left for less, before letting go
Click (Polygon)	Add a corner; double-click or Enter fills it
Right-click (Polygon)	First click: cut the shape out. After: take back a corner
Backspace / Escape	Polygon: take back a corner / give it up
Left-drag	Paint a mask, draw a heading, or measure
[/]	Brush smaller / larger
H (hold)	Look under the masks
Click (Move label)	Put the label there, on its own image
Right-drag	Erase the mask (Paint)
Right-click	Clear the heading (Orient) or the size (Measure)
L / W	Measure: the next line is the length / the width
Ctrl+Enter	Notes: store the description being typed
Enter	Notes: store the Group ID being typed
Escape	Notes: put back what was stored
Shift+drag	Pan the zoomed snippet
Mouse wheel	Zoom to the cursor; kept until Fit
Double-click	Open a grid cell in the Single view
Click (Grid, Review)	Reject the snippet; again takes it back
Right-click (Grid, Review)	Flag the snippet to look at again
A (Grid, Review)	Accept the rest of the page, then the next
Ctrl+Z	Undo - one history with the main window
Ctrl+Y	Redo
Ctrl+S	Save Project

Part 3

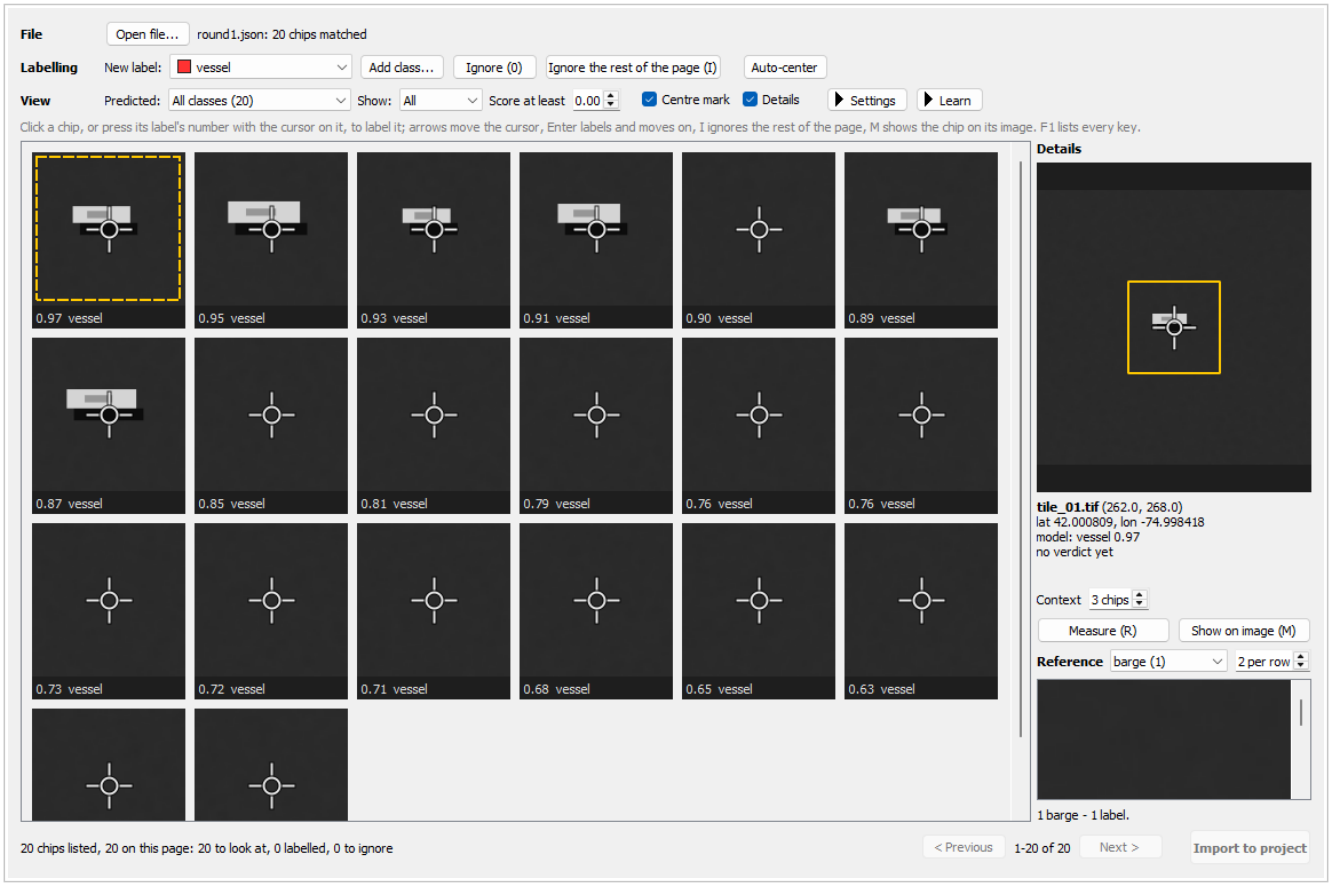
Model Review: what the model got wrong

26 Labelling the model's mistakes

The model team runs their classifier over the imagery in your project and writes out the chips it was confidently wrong about: clutter it took for a vessel, say. Labels > Model Review... opens a window that shows those chips, the surest first, for you to give each its real class - usually a new one, so the model can learn the clutter - or to ignore. One click imports your verdicts into the project as labels. The team retrains, runs again and sends the next file; what you reviewed before stays out of your way.

Opening the team's file

Open file... (Ctrl+O) takes their JSON file. Each record names the chip's image, the image's size, the centre, the class the model predicted and its score. Images are matched by file name, so the team's copy of the imagery can live anywhere. Chips on images that are not in the project are rejected and counted; chips of a class the project does not have are dropped; the line beside the file name says what came in and what did not.



A file of mistakes open: every chip the model called vessel, surest first; the first chip in hand, with its surroundings in the Details panel.

The grid

The controls sit in three rows: FILE, what was opened and what came of it; LABELLING, the verdicts; VIEW, what the grid shows. Two more fold out from the end of the VIEW row: SETTINGS, set once - the chip size, the columns, the chips per page, Show reviewed and Same chip within - and LEARN, the training and suggestion controls described below.

- PREDICTED chooses which of the model's calls to look at - all of them, or one class. Each list runs from the chip the model was surest of, top left, to the one it was least, bottom right; the score is under each chip, and the model's call beside it when the classes are mixed.
- CHIP SIZE is yours, as in the Snippet Editor: the file says only where each centre is. COLUMNS sets how many chips share the width.

- PER PAGE sets how many chips are on the grid at once; Next and Previous, or PageDown and PageUp, turn the pages. A file of thousands opens as quickly as one of ten.
- SHOW lists all of the chips, the ones still to look at, the labelled - all of them, or those labelled one class - or the ignored. The list settles when a page turns, so a chip you have just labelled stays where it is until then.
- SCORE AT LEAST clips the grid to the chips the model scored at least that - 0.95 to tackle its surest mistakes first, 0 for all of them. The status bar counts what it hides; the value is remembered.
- A chip's tooltip names its image and position.

Giving verdicts

- Pick the label in NEW LABEL, or press its number, 1 to 9. Then click a chip to give it that label. Shift-click labels every chip from the last one you clicked to this one.
- Ctrl-click picks chips without labelling them; a number key then labels all the picked at once. Ctrl+A picks every chip on the page, Escape none.
- ADD CLASS... makes a class the project does not have yet - rock, wake, shadow - adds it to the project, and makes it the active label; chips you had picked get it at once.
- 0, or the IGNORE button, is the verdict for a chip that is nothing to label: looked at, remembered, no label made.
- A right-click, or Delete on picked chips, takes a verdict back. A labelled chip wears its class colour and name; an ignored one is grey. Ctrl+Z takes back the last thing you did in the window - a verdict, a run of them, a page's ignores, a centre move - and only then reaches the project's own history.

Working a page from the keyboard

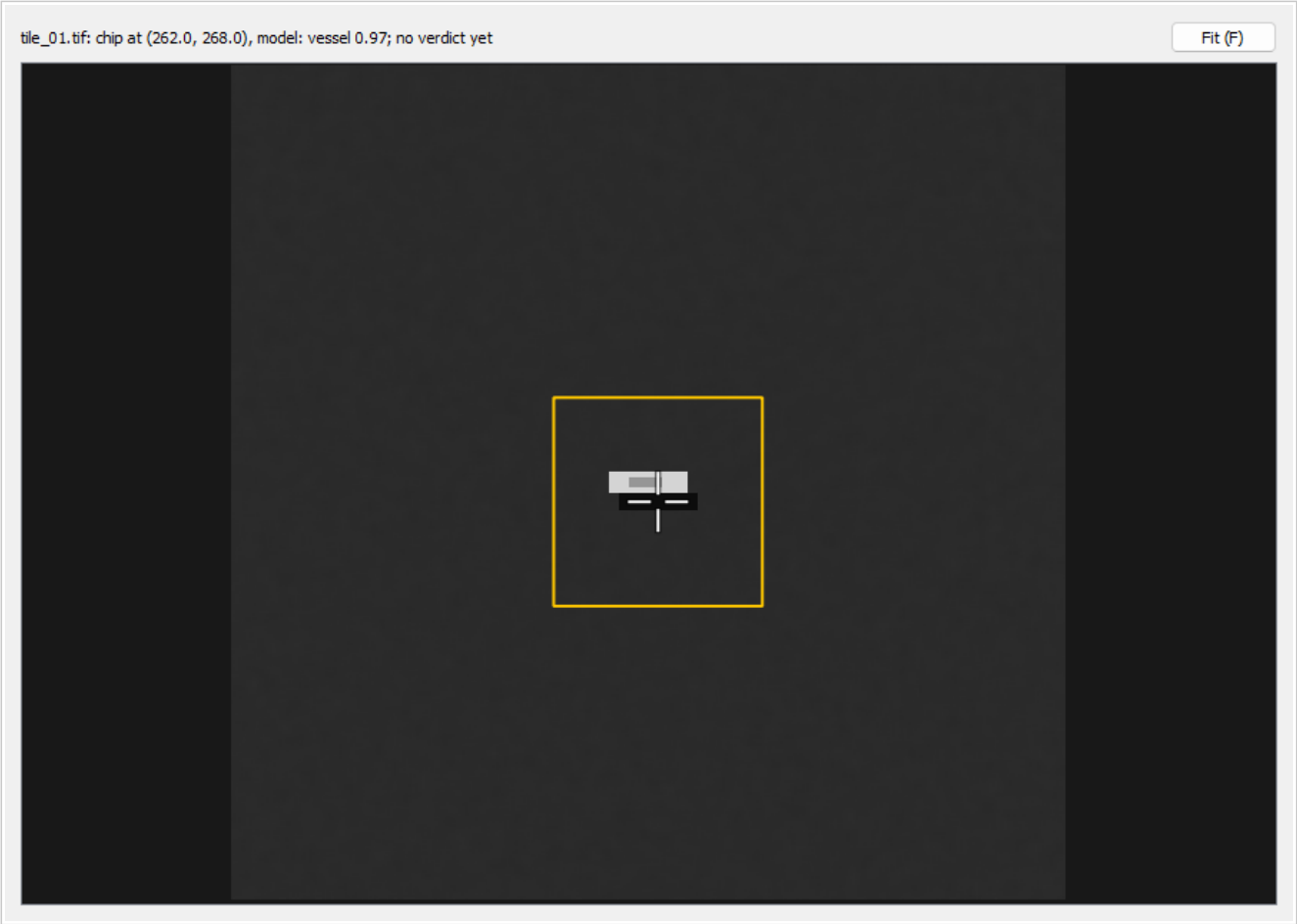
The arrow keys move a dashed cursor from chip to chip; Home and End go to the page's first and last, and stepping past either end turns the page. A number key labels the chip under the cursor and moves on; Enter gives it the label already active and moves on; Space moves on and leaves it as it is; 0 ignores it. When the few chips worth a class on a page are done, I ignores every chip still without a verdict and brings up the next page. Click a chip and the cursor steps aside; the next arrow key puts it after the chip you clicked.

Looking closer

A doubt about a small square usually clears with more of the picture. The DETAILS panel beside the grid shows the chip in hand - the one under the cursor, the one you last clicked, or the one the mouse has come to rest on - read again at CONTEXT times the chip size, with a box where the chip itself is and the mark where the label will go. Under it is everything known: image and position, latitude and longitude, the model's call and score, the verdict, a moved centre, an earlier round's verdict. Under it, REFERENCE shows chips already given a class - this session's verdicts, the newest first, then earlier rounds and the project's labels of that class - to hold up against the chip in hand; pick the class at its head, and how many chips share a row: two by default, one for a single column of big chips. Untick DETAILS to give the grid the whole width.

SHOW ON IMAGE, or M, opens the whole image in a window of its own, with a box the size of the chip where the chip is and the mark at its centre: a shadow beside a hull, a wake, a pattern in the waves explains itself at that scale. The wheel zooms about the pointer, a drag pans, F fits the image again, Escape closes. Press M again and the window follows the chip in hand, keeping the zoom when it is on the same image.

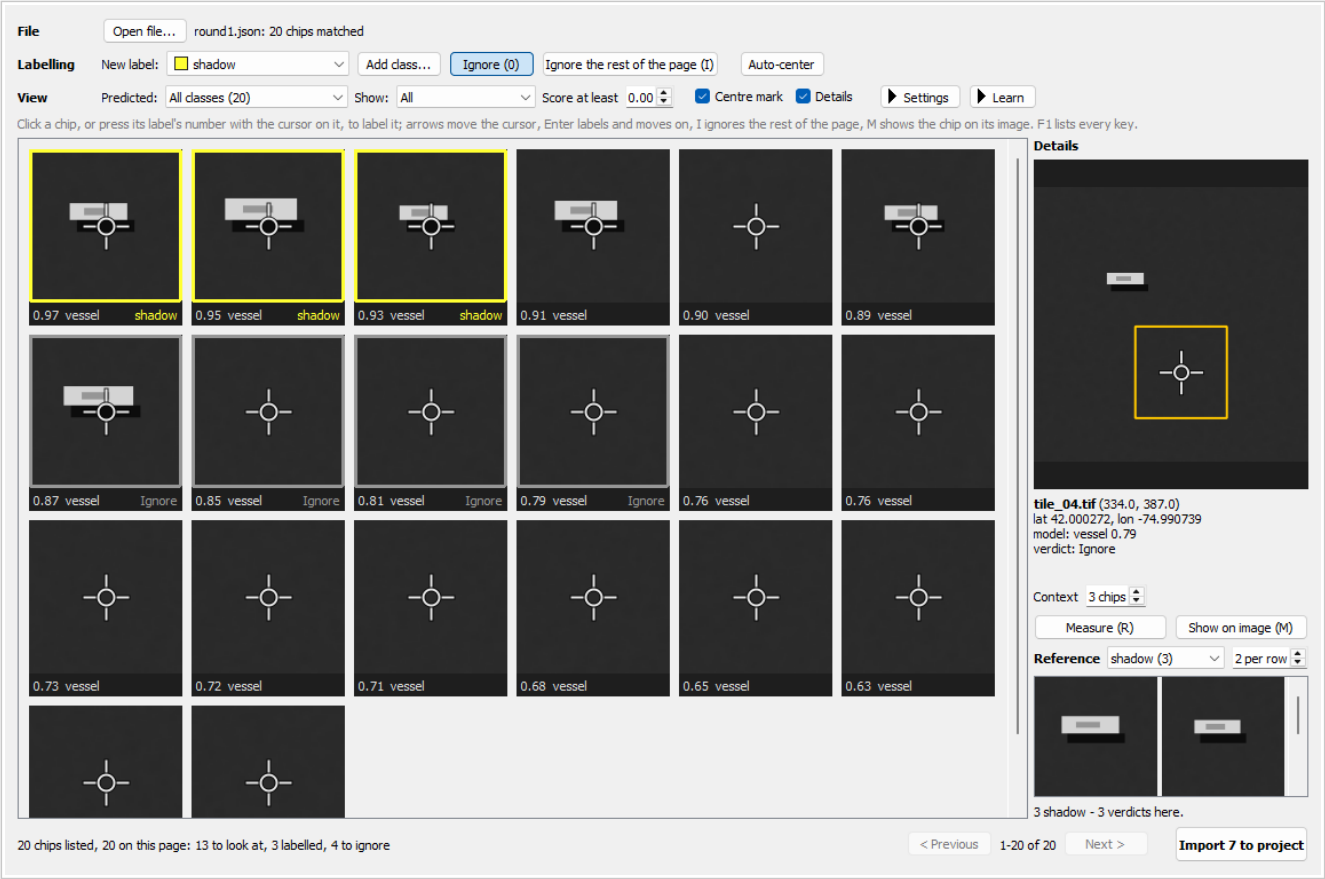
MEASURE, or R, turns the Details picture into a ruler for when the size of a thing decides its class: click two points and the length between them is shown in metres, from the image's georeferencing, and in pixels; an image without georeferencing gives pixels alone. A right-click clears the line, another click starts a new one, and the tool stays on until R is pressed again.



Show on image: the chip's box on the whole of its image.

Putting the centre on the object

The model's centre is where its window sat, which is often beside the clutter rather than on it, and the label would go there. The CENTRE MARK on each chip shows where the label will land. Double-click the object and the centre moves there - without labelling the chip: the chip is read again around the new centre, so the object sits in the middle, and the mark turns cyan. Ctrl+Z takes a centre move back (and then undoes in the project, as always). AUTO-CENTER makes a first guess for the picked chips, or for the whole page with none picked: it moves each centre onto the nearest patch that stands out from the background, bright or dark, and leaves a chip of plain water where it is. Look at where it landed; a double-click corrects it, and a right-click puts the centre back where the project has it - the label's place once imported, else where the file had it - along with taking the verdict back. The memory still matches the next round by the centre the file gave, so a moved label is never mistaken for a new chip.



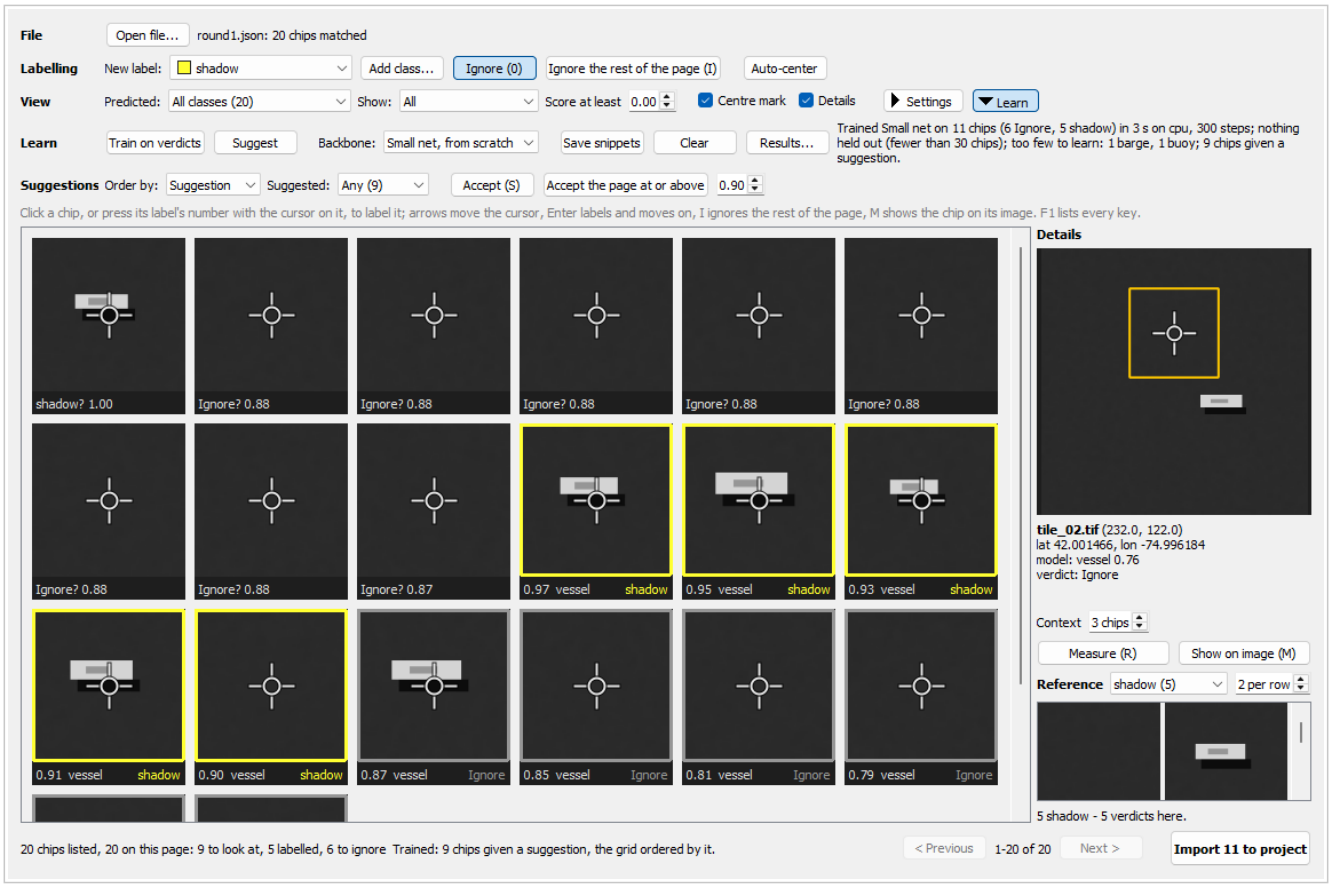
Three shadow chips labelled with a class added on the spot; four water chips given Ignore.

Learning from the verdicts

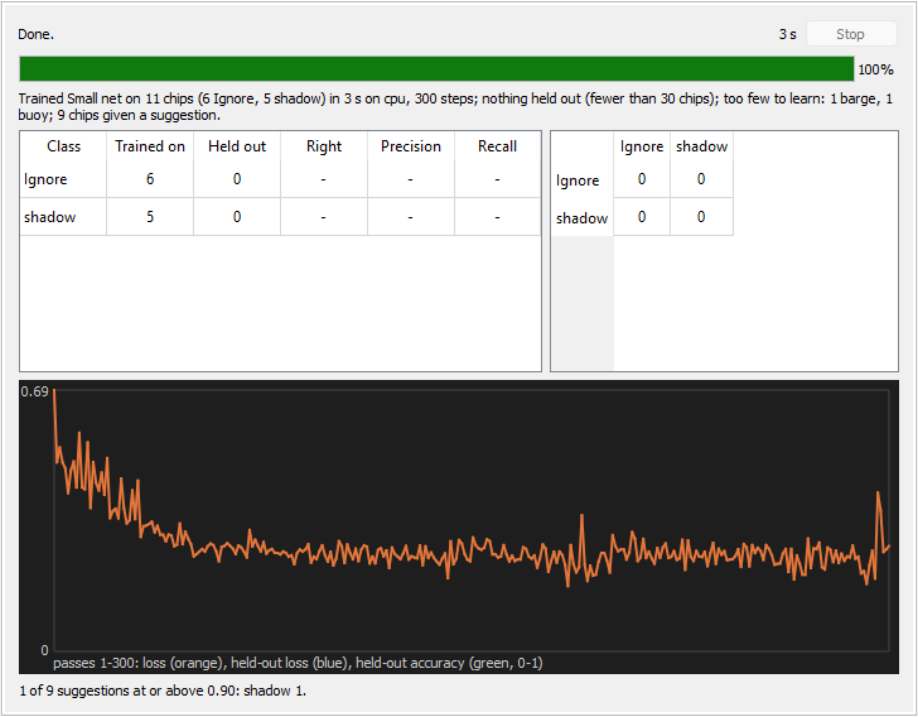
Once a few verdicts are in - five chips of each of two classes, Ignore counting as one - TRAIN ON VERDICTS, under the VIEW row's LEARN button, teaches a model what you have decided: the verdicts on the grid, the project's memory of earlier rounds, and the project's own labels of those classes. BACKBONE chooses the model: a small net learned from scratch, or ImageNet's ResNet18 with a linear head on its features, which does better with few verdicts (it needs torchvision and a 45 MB weights file, fetched once). Training runs in the background, on the CPU or on an NVIDIA GPU when PyTorch was installed for one. A training window shows the phase, a progress bar and Stop, and when the work is done the report: a table per class, the confusion matrix, the loss and held-out accuracy by pass, and how many suggestions the Accept threshold would take; RESULTS... brings it back. From thirty chips a fifth is held out, and the accuracy reported is on those - the number to trust.

Every chip still to look at then shows the model's suggestion in its caption, ROCK? 0.91, and the grid goes over to ORDER BY Suggestion, the surest first; SUGGESTED narrows the grid to one suggested class. S, or ACCEPT, gives the chip under the cursor - or the picked chips - its suggested class; ACCEPT THE PAGE AT OR ABOVE takes every suggestion on the page at least as sure as the box says, as one thing to take back. SUGGEST runs the model again without training - after a chip-size change, or on the next round's file. Train again whenever you like: the model learns from everything decided so far.

Chips are read from the imagery once, scaled small and saved under the system's temporary folder, so training and suggesting read them and not the GeoTIFFs; SAVE SNIPPETS does that ahead of time for the whole file, CLEAR deletes them. Training needs GeoLabeller run from source with PyTorch installed (the README says how); the installed build shows the row greyed out, with the reason.



Trained on five shadows and six water chips: the rest ordered by what the model suggests, surest first.



The training window after a run: the report per class, the confusion matrix, the curves, and what the Accept threshold would take.

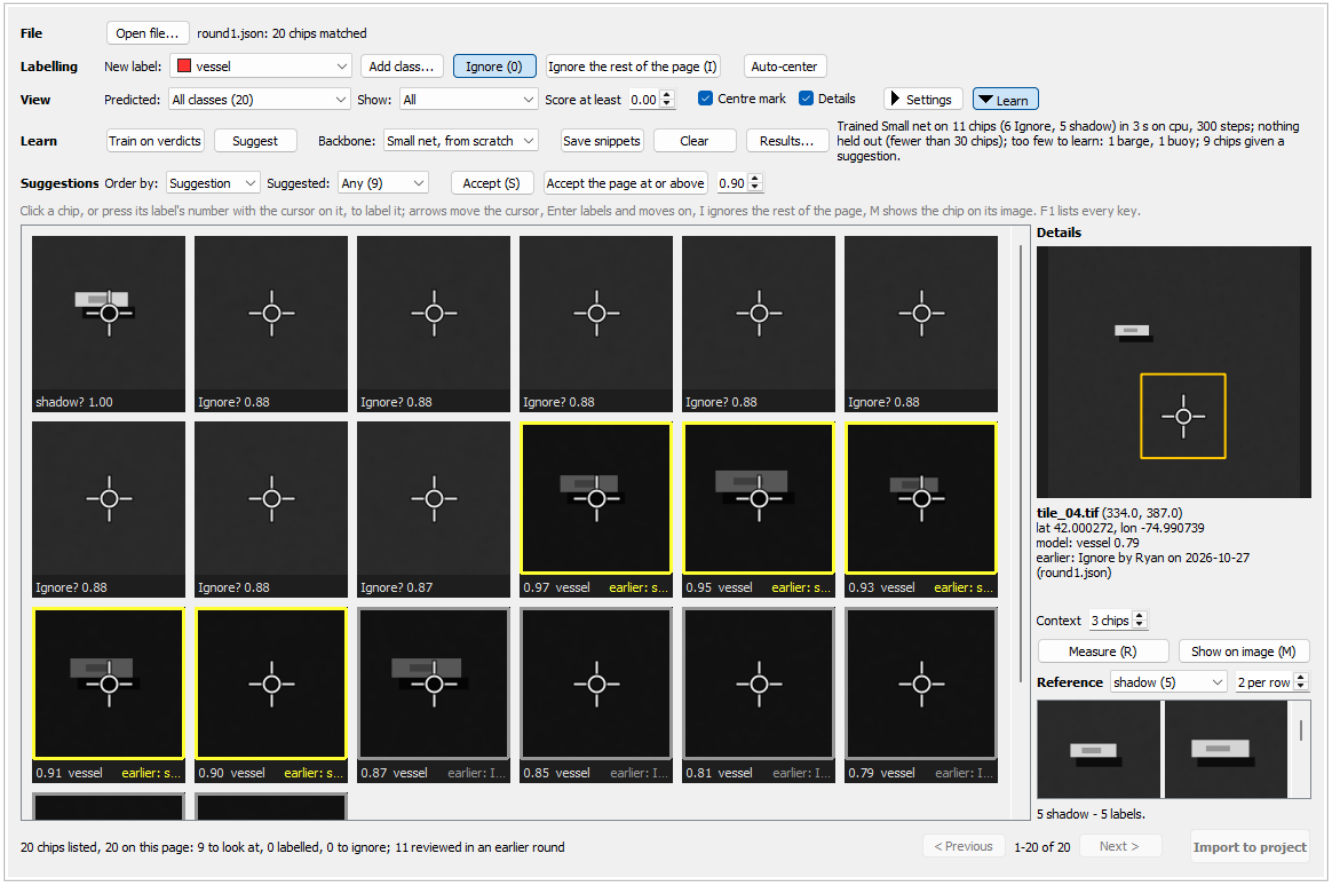
Import to project

IMPORT TO PROJECT, at the bottom right, or Ctrl+Enter, first says what it is about to do - how many chips, how many of each class, how many ignored, how many re-centred - and asks. Then it makes one label per labelled chip, on its image at the chip's centre, with latitude and longitude from the image's georeferencing, your user name, and a

description that says what the model said - for example "model: vessel 0.97, round1.json" - so the label itself tells the story. Ignored chips make no label. It is one undo step in the project. The chips you dealt with leave the grid, the rest wait, and the status line says what came of it.

The next round

The project remembers every verdict, ignores included. When the team's next file arrives, a chip on the same image whose centre lies within a few pixels of one you reviewed is that chip again, and is kept off the grid; SAME CHIP WITHIN sets how close counts, and an object the model flagged under two classes is two chips, each remembered on its own. SHOW REVIEWED brings those chips back, shaded, with what they became, each sitting where its label is. Users make mistakes: give one of them a new verdict - a click, a number, 0 for Ignore - or double-click to move its centre - and the next IMPORT changes the label it made, moves it, removes it for Ignore, or makes one for a chip ignored before; one undo step, as ever. A new verdict waits through anything else you do in the window until that import.



After the import, with Show reviewed ticked: the chips dealt with are shaded, and say what they became.

> Save the project after an import. The memory lives in the project file, and the next round's file is matched against it.

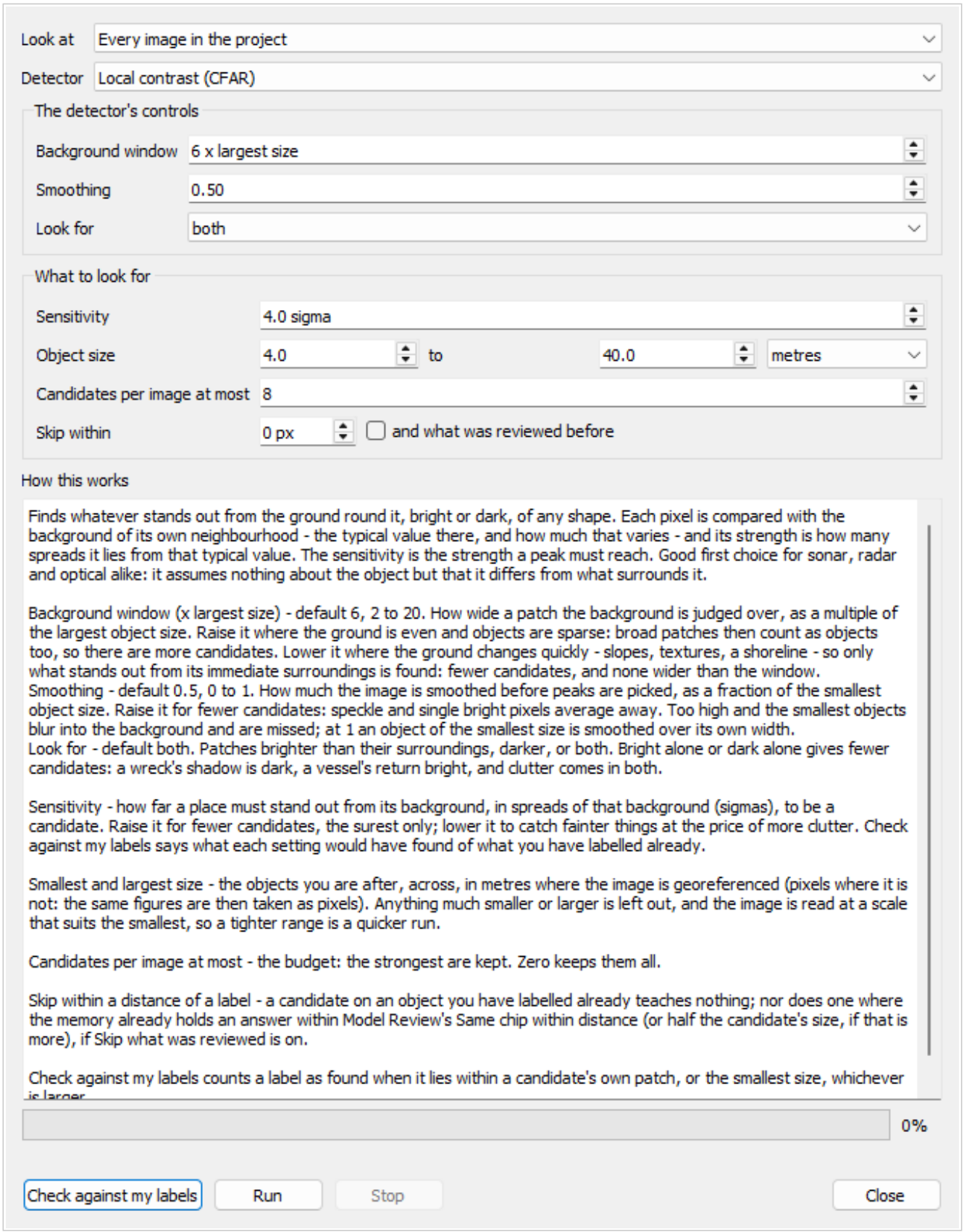
Every key the window takes

Key	Does
Ctrl+O	Open a file of the model's mistakes
Left-click chip	Give it the active label
Shift+Left-click	Label every chip from the last click to this
Ctrl+Left-click	Pick a chip without labelling; a key then labels the picked
1-9	The class at that place in the list: onto the picked chips, or the chip under the cursor (then the next)

0	Ignore: looked at, nothing to label
Arrows / Home / End	Move the cursor from chip to chip; past the page's end it turns the page
Enter	The active label onto the chip under the cursor, then the next
Space	The next chip, this one left as it is
I	Ignore the rest of the page, then the next page
R	Measure: two clicks on the Details picture give the length between them, in metres by the image's georeferencing and in pixels (pixels alone without); a right-click clears
M	Show the chip in hand on its whole image, in a window: the wheel zooms, a drag pans, F fits, Escape closes
Train on verdicts button	A model learned from the verdicts suggests a class for the chips left; Suggest runs it again; Save snippets reads the chips ahead (source runs with PyTorch, requirements-ml.txt)
S	Accept the suggested class: onto the picked chips, or the chip under the cursor (then the next)
Double-click chip	Put the label's centre there, on the object (it does not label)
Auto-center button	Move the picked chips' centres (or the page's) onto what stands out
Right-click / Delete	Take a verdict back, and the centre with it
Escape / Ctrl+A	Pick none / every chip on the page
PageDown / PageUp	Next / previous page of chips
Ctrl+Enter	Import to project
Ctrl+Z	Take back the last thing done here - a verdict, a run, a centre move; then undo in the main window's history
Ctrl+Y	Redo, likewise
Ctrl+S	Save Project

27 Finding candidates with a detector

Nobody has to find the objects by eye first. Labels > Find Candidates... runs a detector over your imagery - any imagery: sonar, radar, optical, one band or three - and proposes the places that stand out from their surroundings, bright or dark, of the sizes you are after. The candidates open in Model Review (chapter 26) exactly as the model team's mistakes do: a grid, surest first, a verdict each, Import makes the labels, and the memory keeps what you dismissed off the grid when you run the detector again. A detector finds what differs from the background; you supply the judgement.



Find Candidates: which images, which detector and its controls, the sizes, the sensitivity, the budget, and the account of how it all works.

Which images, and which detector

LOOK AT chooses the selected group, the images shown on the map, or every image in the project. Two detectors are offered; each documents itself in the HOW THIS WORKS panel, and every control says in its tooltip what it measures and whether raising it gives more candidates or fewer. The same words follow here.

Local contrast (CFAR)

Finds whatever stands out from the ground round it, bright or dark, of any shape. Each pixel is compared with the background of its own neighbourhood - the typical value there, and how much that varies - and its strength is how many spreads it lies from that typical value. The sensitivity is the strength a peak must reach. Good first choice for sonar, radar and optical alike: it assumes nothing about the object but that it differs from what surrounds it.

- Background window (x largest size) - default 6, 2 to 20. How wide a patch the background is judged over, as a multiple of the largest object size. Raise it where the ground is even and objects are sparse: broad patches then count as objects too, so there are more candidates. Lower it where the ground changes quickly - slopes, textures, a shoreline - so only what stands out from its immediate surroundings is found: fewer candidates, and none wider than the window.
- Smoothing - default 0.5, 0 to 1. How much the image is smoothed before peaks are picked, as a fraction of the smallest object size. Raise it for fewer candidates: speckle and single bright pixels average away. Too high and the smallest objects blur into the background and are missed; at 1 an object of the smallest size is smoothed over its own width.
- Look for - default both. Patches brighter than their surroundings, darker, or both. Bright alone or dark alone gives fewer candidates: a wreck's shadow is dark, a vessel's return bright, and clutter comes in both.

Blobs (difference of Gaussians)

Finds compact, round-ish patches of about the sizes you gave, bright or dark against their background. The image is compared with itself blurred at each size between the smallest and the largest; a patch of about that size answers strongly, an edge or a gradient weakly, and speckle hardly at all. The sensitivity is the strength a peak must reach, in spreads of the background. Best for objects of a known size range; a long wake or a ragged patch suits the contrast detector better.

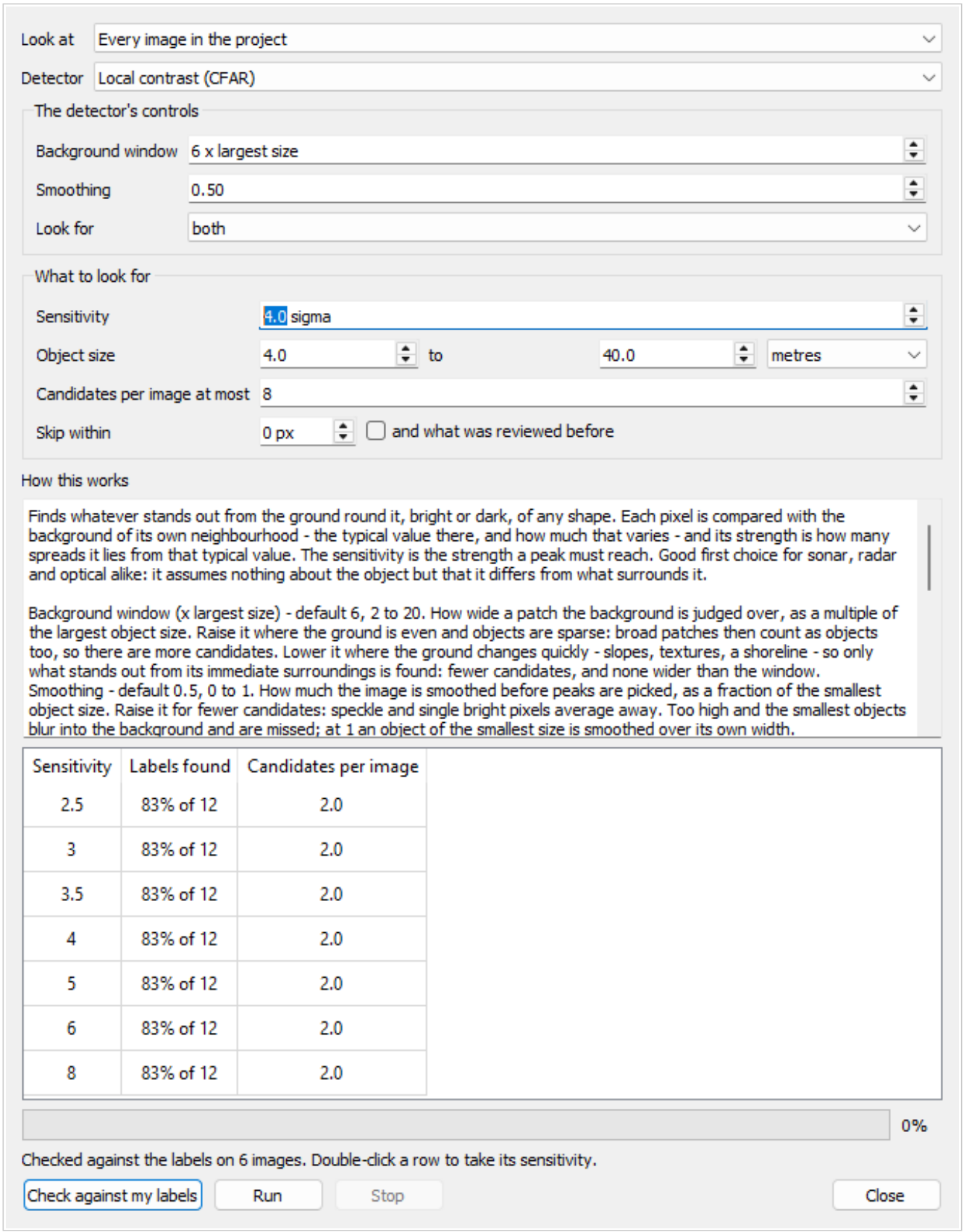
- Sizes per doubling - default 3, 1 to 6. How many sizes are tried between one size and its double, from the smallest to the largest. Raise it for a few more candidates - an object between two sizes answers less to either - at the cost of time: each size is another pass over the image.
- Look for - default both. Patches brighter than their surroundings, darker, or both. Bright alone or dark alone gives fewer candidates: a wreck's shadow is dark, a vessel's return bright, and clutter comes in both.

The controls they share

- Sensitivity - how far a place must stand out from its background, in spreads of that background (sigmas), to be a candidate. Raise it for fewer candidates, the surest only; lower it to catch fainter things at the price of more clutter. Check against my labels says what each setting would have found of what you have labelled already.
- Smallest and largest size - the objects you are after, across, in metres where the image is georeferenced (pixels where it is not: the same figures are then taken as pixels). Anything much smaller or larger is left out, and the image is read at a scale that suits the smallest, so a tighter range is a quicker run.
- Candidates per image at most - the budget: the strongest are kept. Zero keeps them all.
- Skip within a distance of a label - a candidate on an object you have labelled already teaches nothing; nor does one where the memory already holds an answer within Model Review's Same chip within distance (or half the candidate's size, if that is more), if Skip what was reviewed is on.
- Check against my labels counts a label as found when it lies within a candidate's own patch, or the smallest size, whichever is larger.

Checking against your labels

Before anyone reviews a candidate, CHECK AGAINST MY LABELS runs the detector over the labelled images among those chosen and says, for a range of sensitivities, what share of your labels a candidate would have landed on and how many candidates an image would get. That table is how a sensitivity is chosen: double-click a row to take it. Labels already there are not skipped for the check - they are the test.

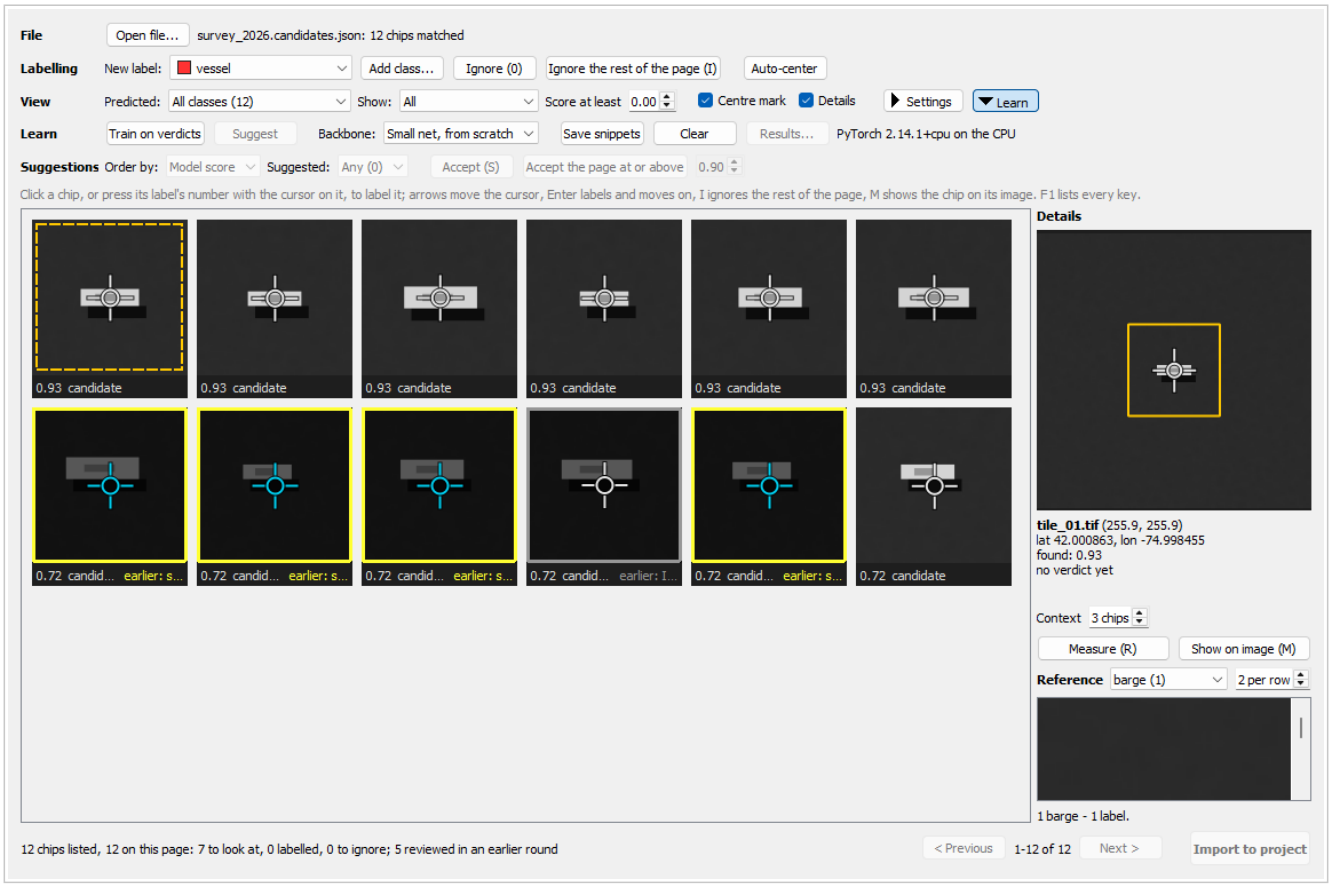


The check: at each sensitivity, the labels found and the candidates per image.

Running, and what happens next

RUN finds the candidates on the images chosen - the window keeps working, with progress and Stop - writes them

to a file beside the project, named after it with .candidates.json, and opens that file in Model Review. The project must be saved first, since the file sits beside it. Candidates on labels already there, and where the memory already holds an answer, are left out and counted; so are those smaller or larger than the sizes given, and all but the strongest few on an image when a budget is set. The file is an ordinary Model Review file, so Open file... takes it again later.



The candidates in Model Review: one group, "candidate", surest first, the first in hand with its surroundings.

In the grid each candidate reads "found" and its score - how far it stood out, half at eight sigmas - where a model's chip reads its class and likelihood. Give each its class or Ignore it, as in chapter 26, and Import. Run the detector again after more labelling, or at another sensitivity, and only what is new comes up: the labels and the memory keep the rest away.

> Start with a tight size range and the check. A range that fits the objects you want is a quicker run and a cleaner grid; the check tells you what a sensitivity costs before you spend a review on it.

Every key the window takes

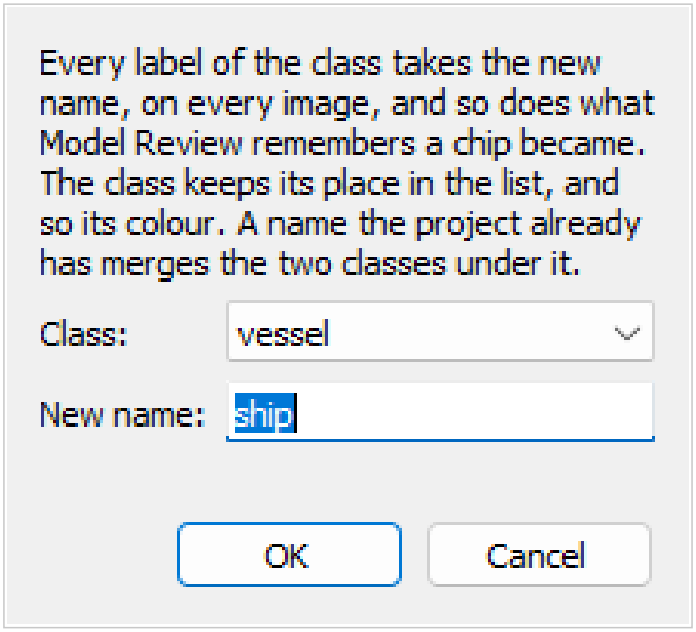
Key	Does
Labels > Find Candidates...	A detector proposes places that stand out, to check in Model Review
Check against my labels	What each sensitivity would have found of the labels, and how many candidates an image gets
Run	Write the candidates file beside the project and open it in Model Review
Double-click a row	Take that sensitivity

28 Keeping labels tidy

Four small tools for a project that has been labelled for a while: a class renamed without losing its labels, the labels that landed on top of each other thinned, an object seen in a dozen images drawn as one marker, and a label's snippet at a glance.

Renaming a class

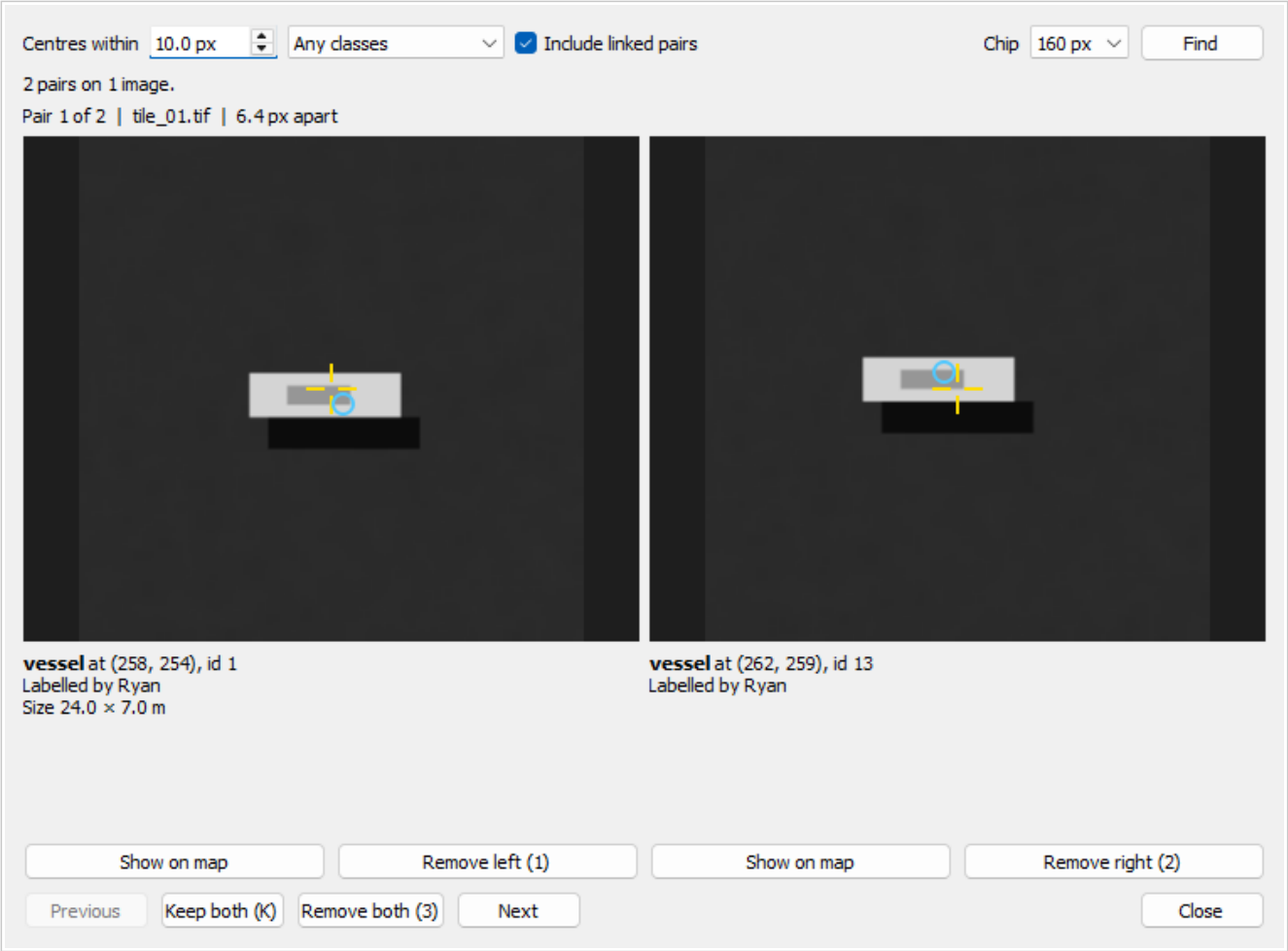
Edit Classes reads a changed line as one class removed and another added - and removing a class deletes its labels. To rename a class and keep them, use LABELS > RENAME CLASS...: pick the class, type the new name, OK. Every label of the class on every image takes the name, and so does what Model Review remembers a chip became; the class keeps its place in the list, and so its colour. One undo step. A name the project already has is a merge - the dialog says so, and the window asks, with the counts, before the two become one class.



Rename Class: the class, its new name, and what will happen.

Cleaning close labels

Labels pile up on an object: two people labelled it, an import kept both, a hand slipped. LABELS > CLEAN CLOSE LABELS... finds, on every image, each two labels whose centres lie within the distance you give (in pixels of the image; the Import Options distance to start with) - any classes, one class only, or only different classes, with or without pairs that are one linked object already - and shows them one at a time: each label's snippet beside the other's, a cross where the label is and a ring where its partner lies, the class, position, description and labeller under each, and SHOW ON MAP for either.



A pair, 4 px apart on one tile: the two snippets, and what to do with them.

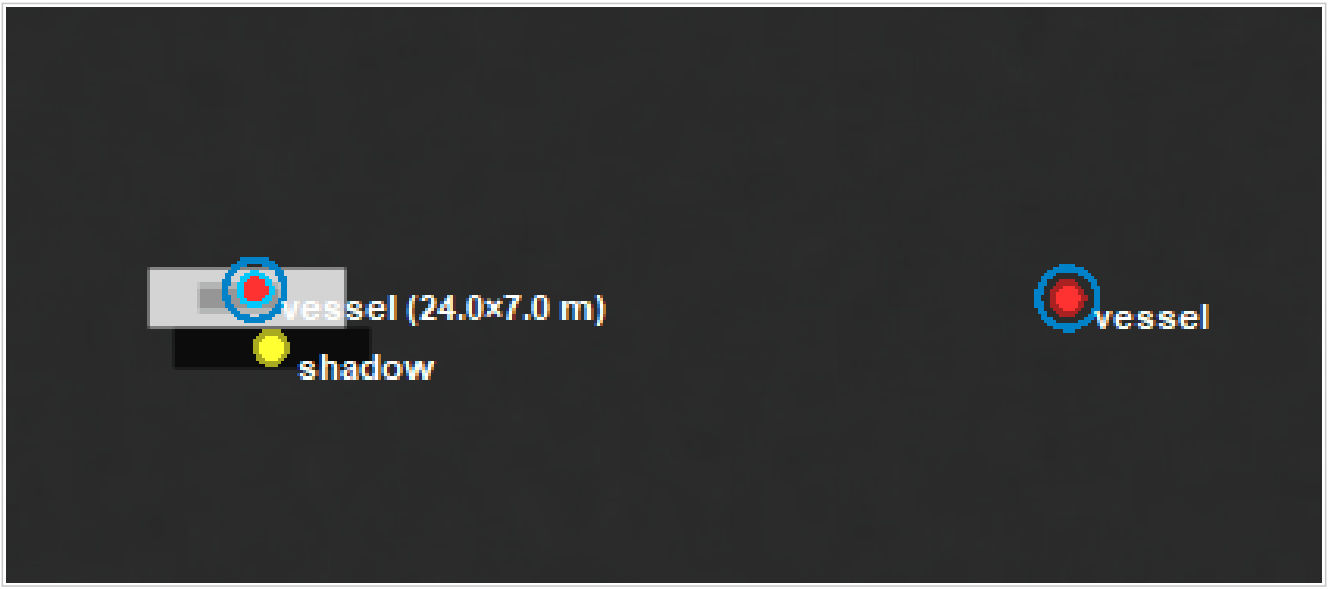
- KEEP BOTH (K) leaves the two as they are and goes on; REMOVE LEFT (1), REMOVE RIGHT (2) and REMOVE BOTH (3) remove at once, each one undo step of the main window's history (Ctrl+Z there takes it back). The arrows move between pairs.
- Every pair that involved a removed label goes with it, so a run of three close labels is never offered against a label that is gone. The survivor of a pair keeps its links; nothing here edits an object.
- The distance, the class filter, the linked choice and the chip size are remembered.

One marker per object

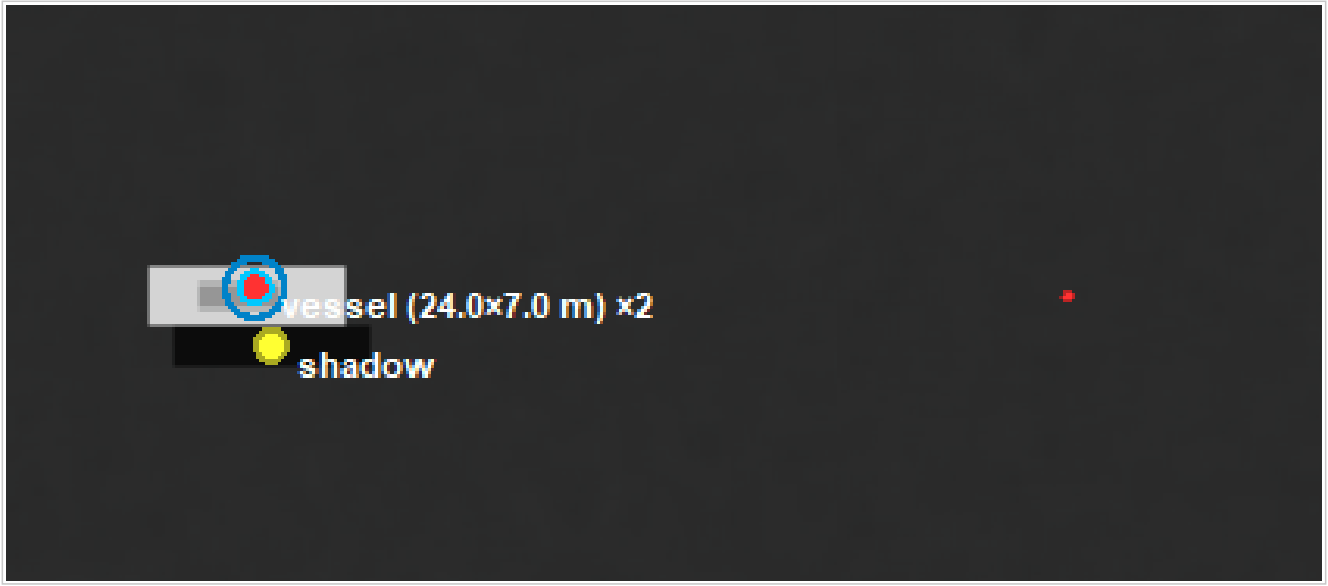
An object seen in a dozen images wore a dozen circles on the map, and the object was somewhere under them. VIEW > LABEL MARKERS chooses how the labels of one linked object are drawn:

- EVERY LABEL - as always: each label its own marker.
- ONE PER OBJECT, THE REST AS DOTS - one marker for the object, its caption counting the views (vessel ×2); the other labels are small dots, still there to right-click.
- ONE PER OBJECT - the one marker alone; the other labels are not drawn until their image is the one a cycle is showing, or one is picked from the Labeled panel or the snippet sidebar.

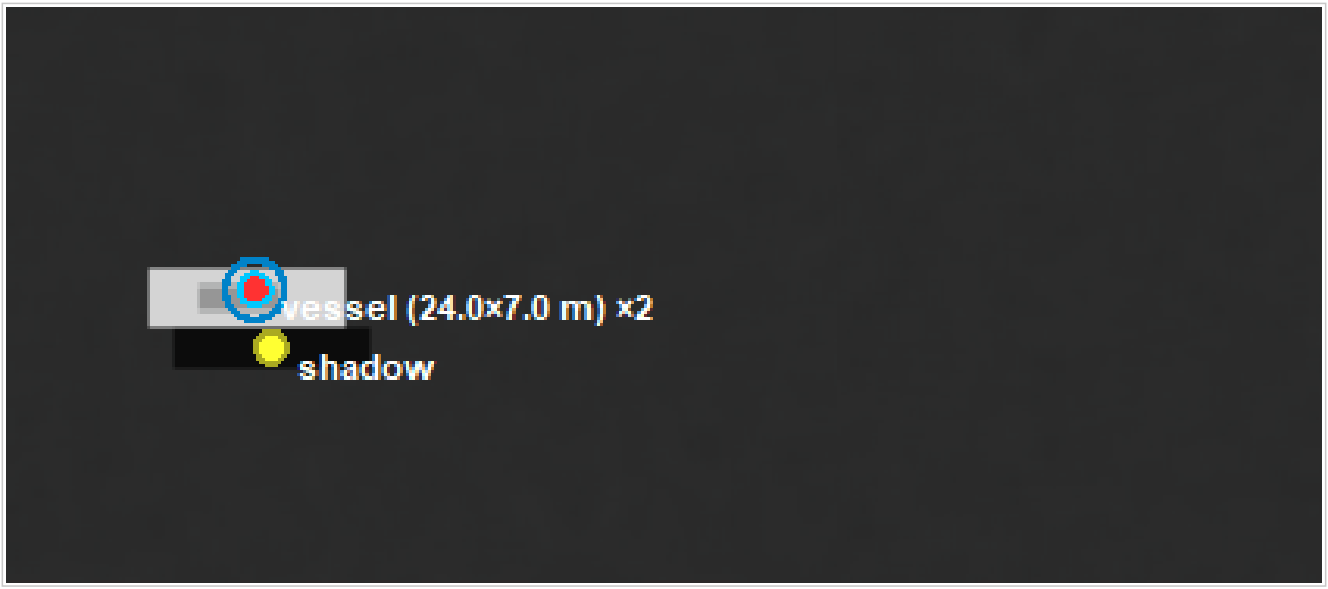
The marker drawn is the label on the image being shown; failing that, the one you asked to see; failing that, the one on the topmost layer that is switched on. The choice is remembered.



Every label: the vessel's two markers, one from each tile, with their halos.



The rest as dots: one marker, counting two views, and a dot where the other view is.



One per object: the marker alone.

A snippet on hover

Hold CTRL and rest the mouse on a label's marker: a small window shows the snippet around that label, drawn the way the group's Display Settings draw it, with its class, image and description - a quick look without opening the Snippet Editor. It goes when the mouse leaves the marker, Ctrl is released, or anything is clicked. Only with Ctrl held, so a mouse crossing the map while you pan raises nothing. VIEW > HOVER SNIPPET (HOLD CTRL) sets the size - how many source pixels are read, and how big the window is - or Off; the choice is remembered.



The hover snippet: the vessel at 128 px, with its class and tile.

Every key these take

Key	Does
Labels > Rename Class...	A new name for a class on every label that has it; a name already there merges the two
Labels > Clean Close Labels...	Every two labels within a distance on one image, side by side: keep both, remove one, or both
Left / Right	Clean Close Labels: the previous / next pair
K	Clean Close Labels: keep both, then the next pair

1 / 2 / 3	Clean Close Labels: remove the left / the right / both (one undo step each)
View > Label Markers	A linked object as one marker (counting its views), the other views as dots or not drawn; or every label
Ctrl + rest on a label	Its snippet beside the cursor (View > Hover Snippet sets the size, or Off)